Lorentzian propinquity on truncated $\mathrm{SU}(2) \times U(1)_T$ Krein spectral triples: a degeneracy theorem

J. Loutey*

June 2026 (preprint)

Abstract

We construct truncated Krein spectral triples over the compact-temporal Wick-rotated manifold $S^3 \times S^1_T = S^3 \times S^1_T$: $\mathcal{K}_{n_{\max},N_t} := \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ is the chirality-doubled Camporesi–Higuchi spinor space tensored with the N_t-mode Fourier truncation of $L^2(S^1_T)$, with fundamental symmetry $J_L = J_L^{\mathrm{spat}} \otimes I_{N_t}$ at the Bizi–Brouder–Besnard [3] classification $(m,n) = (4,6)$, Lorentzian Dirac $D_L = i(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t})$ per van den Dungen 2016 [32] Proposition 4.1, and operator system built from chirality-doubled scalar 3-Y spatial multipliers tensored with momentum-diagonal temporal multipliers. The natural device for a "Lorentzian propinquity" on this family is to compress to the Krein-positive subspace $\mathcal{K}^+$, where the Krein product is a positive-definite Hilbert inner product, and to apply Riemannian quantum-metric machinery to the compressed triple $(P_+\mathcal{O}^L P_+, \mathcal{K}^+, P_+ D_L P_+)$. Our main theorem shows that this device fails structurally (Theorem 5.2): Krein-self-adjointness of the anti-Hermitian spatial block forces $\{J_L, D_{\mathrm{GV}} \otimes I_{N_t}\} = 0$, so the compression annihilates the spatial Dirac, $P_+(D_{\mathrm{GV}} \otimes I)P_+ = 0$, while the momentum-diagonal temporal multipliers commute with the surviving temporal block; the compressed-Dirac Lipschitz seminorm therefore vanishes *identically* on the operator system (verified bit-exactly at $(n_{\max}, N_t) \in \{(2,3),(3,5)\}$: kernel = 100% of the multiplier basis), and no quantum Gromov–Hausdorff-type distance is defined. What does hold: the operator-level J_L-equivariance of the Berezin/truncation pair; the structural identity $[D_L, a_s \otimes a_t] = i[D_{\mathrm{GV}}, a_s] \otimes a_t$, which is precisely the diagnosis that the momentum-diagonal temporal algebra is metrically invisible; and a spatial convergence statement (Proposition 5.4): the spatial truncations converge in van Suijlekom's state-space Gromov–Hausdorff distance [10, 33] at the companion-paper rate $O(\log n_{\max}/n_{\max})$ — unconditional, via the companion paper's translation-seminorm theorem — with the temporal factor a metrically invisible spectator carried J_L-equivariantly. The genuine Lorentzian/Krein quantum-metric convergence problem — a temporal multiplier algebra the Dirac can see (Toeplitz compressions of e^{iqt}) and a Krein-compatible Lipschitz seminorm that does not compress D_L — is stated as an open target with a sharpened specification (§ 7.3). An earlier draft of this paper claimed a convergence theorem through the compression device; that claim is retracted (Remark 1.1).

MSC2020 classifications: 58B34 (noncommutative geometry), 46L07 (operator systems), 46L60 (applications of selfadjoint operator algebras to physics), 46L87 (noncommutative Riemannian and metric geometry), 46C20 (spaces with indefinite inner product (Krein spaces)).

Keywords: quantum Gromov–Hausdorff convergence, Krein spectral triple, Bizi–Brouder–Besnard indefinite signature, Camporesi–Higuchi Dirac, spectral truncation, central spectral Fejér kernel, Berezin reconstruction, Krein-positive state space, Lipschitz-seminorm degeneracy.

*Independent researcher. `jloutey@gmail.com`. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude. Implementation, internal memos, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author's responsibility.

1 Introduction

This paper concerns the convergence-theorem leg of the Lorentzian-extension programme for truncated spectral triples of the Geometric Vacuum (GeoVac) framework [36] — and proves that the leg's most natural device fails structurally — following the math.OA quartet Papers 38–40 [37–39] (Riemannian propinquity convergence on SU(2), on tensor products of SU(2) triples at distinct focal lengths, and on general compact Lie groups with bi-invariant metric) and the Lorentzian extensions Papers 42–44 [40–42] (Tomita–Takesaki modular structure on Riemannian truncations; four-witness Wick-rotation at the operator-system level on the Lorentzian Krein wedge; operator-system substrate of the Lorentzian Krein spectral triple at finite cutoff).

The dominant blocker for transferring quantum Gromov–Hausdorff convergence [16, 17, 33] to the Lorentzian setting is the absence of a published metric on Krein-signature spectral triples in their own right: the Riemannian constructions crucially use the operator-norm Lipschitz seminorm via the Dirac commutator $\|[D,\cdot]\|_{\mathrm{op}}$, well-defined on Hilbert space but requiring an indefinite-inner-product analog on a Krein space (with the appropriate Krein-self-adjointness of D_L per van den Dungen 2016 [32] / BBB [3]). The natural attempt to sidestep this gap is to compress to the Krein-positive subspace $\mathcal{K}^+$, where the Krein product reduces to a positive-definite Hilbert inner product, and to apply the Riemannian machinery to the compressed triple — the $\mathcal{K}^+$*-restricted weak-form* device (Definition 2.1). The central result of this paper is that the sidestep fails structurally (Theorem 5.2): Krein-self-adjointness forces the compression to annihilate the spatial Dirac, and the resulting Lipschitz seminorm vanishes identically on the operator system. The gap is real, and the repaired target is stated in § 7.3.

1.1 What this paper does

We work on the compact-temporal Wick-rotated geometry $S^3 \times S^1_T$, with Riemannian metric $ds^2 = d\chi^2 + \sin^2\chi\, d\Omega^2 + d\tau^2$ at canonical circumference $T = 2\pi$. The Krein structure J_L is preserved as an *algebraic* object (operator system plus symmetry, in the BBB $(4, 6)$ classification) but no Lorentzian-causal-structure claim is made; the closed timelike-curve obstruction (Geroch [11]) is acknowledged explicitly via the Wick-rotated convention. The paper's content is organized as follows:

(L1′) The Lorentzian operator-system substrate $\mathcal{O}^L_{n_{\max},N_t,T}$ of Paper 44 [42], with propagation number 2 on the achievable envelope and trivial Krein-positivity at the operator-multiplier level.

(L2) A joint central spectral Fejér kernel $K^{\mathrm{joint}} = K^{\mathrm{SU(2)}}_{n_{\max}} \otimes K^{U(1)}_{N_t,T}$ on $\mathrm{SU(2)} \times U(1)$. The smoothing map is unital completely positive with $\|S_{K^{\mathrm{joint}}}\|_{\mathrm{cb}} = 1$; the value $2/(n_{\max}+1)$ is the maximum of the SU(2) Plancherel *mass distribution*, a distinct object (Lemma 4.1).

(L3) The joint Lipschitz comparison $\|[D_L, a]\|_{\mathrm{op}} \le C_3^{\mathrm{joint}} \|\nabla^{\mathrm{joint}} a\|_\infty$ with $C_3^{\mathrm{joint}} \le 1$, via the structural identity $[D_L, a_s \otimes a_t] = i[D_{\mathrm{GV}}, a_s] \otimes a_t$ — read correctly, this identity is the *diagnosis* that the temporal factor is metrically invisible to the seminorm, not a simplification.

(L4) A joint Berezin map $B^{\mathrm{joint}}_{n_{\max},N_t,T}$ (convolution form) with positivity, contractivity, approximate identity (rate $\gamma^{\mathrm{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$), and spatial L3 compatibility; its temporal image is the Toeplitz compression algebra, not the momentum-diagonal subalgebra of $\mathcal{O}^L$ (Remark 4.9).

(Main) The annihilation theorem (Theorem 5.2), an anatomy of why the natural tunneling-pair assembly produces no distance bound (§ 5.3), and the spatial convergence statement (Proposition 5.4), unconditional via the companion paper's translation-seminorm theorem.

Remark 1.1 (History and retraction)**.** An earlier draft of this paper (May 2026; archived in the repository history and the corresponding Zenodo deposit) claimed a "$\mathcal{K}^+$-restricted weak-form Lorentzian propinquity" convergence theorem through Definition 2.1, resting on theorem numbers that do not exist in the cited source [16] and on a cb-norm statement that conflated the kernel's mass distribution with its multiplier symbol. A verification pass (2026-06-09; audit record in the repository `CHANGELOG`, v3.106.0) found the degeneracy proved in Theorem 5.2; the convergence claim is retracted. Dependent claims in the companion papers carry Status notes; the companion S^3 paper's spatial convergence statement survives in corrected form, now unconditional via its translation-seminorm theorem.

1.2 Honest scope and named gaps

The $\mathcal{K}^+$-preservation algebra holds at the operator level (Remark 3.3) — but it preserves a structure whose Lipschitz seminorm is identically zero (Theorem 5.2), so there is no metric content to preserve. The named gaps:

(G1) *Lorentzian/Krein quantum-metric convergence.* A quantum-metric structure on Krein-signature spectral triples is not in the published literature as of June 2026, and the present paper shows that the obvious $\mathcal{K}^+$-compression device cannot supply it: the compressed Dirac loses its entire spatial part by Krein-self-adjointness. Two ingredients are jointly required: a temporal multiplier algebra the Dirac can see (Toeplitz compressions of multiplication operators e^{iqt}, in the Connes–van Suijlekom S^1 pattern [6], rather than momentum-diagonal polynomials), and a Krein-compatible Lipschitz seminorm that does not compress D_L. This is the reopened named research target (§ 7.3, Q1).

(G2) *De-compactification limit $T \to \infty$.* The norm-resolvent / spectral level of this limit is closed in Paper 47 [44] § 4–5 (outer arrow: norm-resolvent convergence of the Lorentzian Dirac on $S^3 \times S^1_T$ to $S^3 \times \mathbb{R}_t$ with exponential tail $O(e^{-|\operatorname{Im} z|\, T/2})$ on compact-support subspaces); that arrow is operator-level, does not use the Lipschitz seminorm, and is unaffected by the degeneracy. The metric level of the limit is open, downstream of G1 (a "zero-cost temporal extent element" through the degenerate quantity does not close it).

(G3) *Cross-manifold extensions: CLOSED as structural impossibility.* Paper 32 Theorem 7.4 proves that the tensor product $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ cannot preserve both the Riemannian-Dirac character of S^3 and the π-free Bargmann character of S^5 within the standard NCG framework. The obstruction is categorical (Bertrand + Fock/HO rigidity), propagating to five structural layers (Paper 24 [35] § V).

(G4) *Inner-factor calibration data* (the Higgs / Yukawa selection question of the Connes–Marcolli SM construction) is orthogonal to the present convergence theorem.

1.3 Related work

Two adjacent Riemannian convergence frameworks exist. Latrémolière's quantum Gromov–Hausdorff propinquity [17] (Trans. AMS 2016) and its metric-spectral-triple extension [16] (Adv. Math. 2022) define convergence via tunnels carried by quantum metric spaces with quantum isometries; the most recent continuity result in that line is Farsi–Latrémolière 2025 [8], treating analytic paths of Riemannian metrics. Separately, van Suijlekom's state-space Gromov–Hausdorff framework [33] bounds the distance between truncated and full state spaces via approximation maps in both directions; this is the framework the constructions of this paper (and of the companion Papers 38–40) actually instantiate — Latrémolière's propinquity imposes tunnel hypotheses not available on these truncations. None of these constructions extend to Krein signature.

Connes–van Suijlekom 2021 [6] introduces the operator-system viewpoint on spectral truncations and proves on the circle S^1 that the truncated state-space distance dominates the Kantorovich metric, deferring Gromov–Hausdorff convergence ("reported elsewhere"); the convergence itself is established in van Suijlekom [33] and extended to all compact metric groups by Gaudillot-Estrada–van Suijlekom [10]. The adjacent program [13, 14, 18, 30] covers $\mathbb{T}^d$, truncated circle geometry, noncommutative integrals on truncations, and groups with polynomial growth, with the $4/\pi$ rate asymptote on compact Lie groups computed in Paper 40 [39] (also adapted to SU(2) in Paper 38 [37]; both with the mass-distribution/multiplier-symbol distinction of Lemma 4.1). All of these are strictly Riemannian.

The Lorentzian Krein-spectral-triple framework derives from Strohmaier [28] (axiomatic foundation), Franco–Eckstein [9] (algebraic causality), van den Dungen 2016 [32] (Krein spectral triples and fermionic action; the Wick-rotation lift used here), Bizi–Brouder–Besnard 2018 [3] ((m, n) classification of indefinite signatures used at $(4, 6)$ West-coast in the present paper), Devastato–Lizzi–Martinetti [7], and the recent SU(1,1) construction [34]; none of these works define a metric on the algebra of operator systems extracted from a Krein triple. The closest algebraic alternative is the Nieuviarts twist morphism [22, 23], which produces pseudo-Riemannian spectral triples from Riemannian ones via a twist on twisted spectral triples; this is not a metric and moreover (as noted in Paper 42 § 3.2) is restricted to even-dimensional manifolds, so does not apply to S^3 at KO-dimension 3.

The adjacent synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämann [20] extends Minguzzi–Suhr's earlier work [19] to Lorentzian pre-length spaces via ϵ-nets of causal diamonds; a downstream pure-metric extension via the intrinsic timed-Hausdorff distance appears in Che–Perales–Sormani [21]. This is conceptually adjacent (all are "Lorentzian Gromov–Hausdorff-type convergence") but lives on a categorically different mathematical object: pre-length spaces with causal-diamond / timed-Hausdorff geometry, not operator-algebraic spectral triples. The two families of constructions are not in competition; neither implies the other.

1.4 Outline

§ 2 fixes conventions: Krein space, Lorentzian Dirac, operator system, joint Fejér kernel, Krein-positive subspace, and the definition of the $\mathcal{K}^+$-restricted weak-form propinquity. § 3 reviews the L1′ operator-system substrate (Paper 44). § 4 treats the three analytical lemmas: L2 (smoothing map and mass distribution, § 4.1), L3 (joint Lipschitz comparison, § 4.2), and L4 (joint Berezin map, convolution form, § 4.3). § 5 proves the annihilation theorem, dissects the failure of the compression assembly, and states the spatial convergence statement. § 6 presents the rate-formula panel and the $N_t = 1$ reduction. § 7 discusses the four-witness Wick-rotation connection (Paper 42/43), states the reopened open questions, and positions the work against Mondino–Sämann. Appendix A records the symbolic Plancherel calculation for the joint Fejér symbol. Appendix B records the numerical panel data.

2 Preliminaries

2.1 Compact-temporal manifold and Wick-rotated metric

The compact-temporal Lorentzian / Wick-rotated manifold is $S^3 \times S^1_T$, with $S^3 = \mathrm{SU}(2)$ the round unit three-sphere (spatial slot) and $S^1_T = \mathbb{R}/T\mathbb{Z}$ the temporal circle of circumference T. The Riemannian / Wick-rotated metric is

$$ds^2 \;=\; d\chi^2 + \sin^2\chi \, d\Omega_2^2 + d\tau^2, \tag{1}$$

with $\chi \in [0, \pi]$ the polar angle on S^3, $d\Omega_2^2$ the round metric on the $\mathbb{S}^2$ base of the Hopf fibration, and $\tau \in \mathbb{R}/T\mathbb{Z}$ the temporal coordinate. We treat the Krein structure J_L defined below as an *algebraic* object that encodes the BBB $(4,6)$ chirality grading without claiming a Lorentzian-causal interpretation of the underlying metric: Geroch's no-CTC obstruction [11] prevents a Lorentzian metric on $S^3 \times S^1_T$ in which S^1_T is timelike, so the present construction operates on the Wick-rotated geometry and lifts the Krein structure as indefinite-inner-product datum (per Strohmaier [28], vdD [32], BBB [3]).

The canonical choice $T = 2\pi$ matches the Bisognano–Wichmann [2] modular period at unit surface gravity; this is consistent with the four-witness arc closure of Paper 43 [41]. The proof of the main theorem holds for any $T > 0$; explicit dependence on T enters the $U(1)$ factor of the joint mass-concentration moment as $O(T/N_t)$.

2.2 Krein space and fundamental symmetry

Per Paper 43 [41]§ 2 / Paper 44 [42]§ 2, the Krein space is

$$\mathcal{K}_{n_{\max},N_t} \;:=\; \mathcal{H}^{n_{\max}}_{\mathrm{GV}} \otimes \mathbb{C}^{N_t}, \tag{2}$$

where $\mathcal{H}^{n_{\max}}_{\mathrm{GV}}$ is the chirality-doubled Camporesi–Higuchi [5] spinor Hilbert space truncated at the $n_{\max}$-th Fock shell, of dimension

$$\dim \mathcal{H}^{n_{\max}}_{\mathrm{GV}} \;=\; \tfrac{2}{3}\, n_{\max}(n_{\max}+1)(n_{\max}+2), \tag{3}$$

and $\mathbb{C}^{N_t}$ is the N_t-mode Fourier truncation of $L^2(S^1_T)$. The fundamental symmetry is

$$J_L \;=\; J_L^{\mathrm{spat}} \otimes I_{N_t}, \tag{4}$$

with $J_L^{\mathrm{spat}} = \gamma^0$ the chirality-swap in the Peskin–Schroeder chiral basis, $(J_L^{\mathrm{spat}})^2 = +I$.

The BBB (m,n) classification at signature $(s,t) = (3,1)$ West-coast assigns $(m,n) = (s+t, t-s \bmod 8) = (4,6)$ via BBB [3] Table 3, fixing $\epsilon = +1, \epsilon'' = -1, \kappa = -1, \kappa'' = +1$. These signs were verified bit-exact at finite cutoff in Paper 43 § 5 [41].

2.3 Lorentzian Dirac operator

The Lorentzian Dirac, per vdD 2016 [32] Prop. 4.1 applied at signature $(3,1)$, reads

$$D_L \;=\; i\left(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t}\right), \tag{5}$$

with ∂_t Fourier-diagonal anti-Hermitian on $\mathbb{C}^{N_t}$: in the momentum eigenbasis $\{|k\rangle : k \in \mathbb{Z}, |k| \leq K_{\max}\}$ with $K_{\max} := \lfloor (N_t-1)/2 \rfloor$ for odd N_t or $N_t/2$ for even N_t,

$$\partial_t \;=\; i\,\mathrm{diag}\big(\omega_k\big), \qquad \omega_k = 2\pi k/T, \tag{6}$$

so $\partial_t^\dagger = -\partial_t$. The spatial part D_{GV} is the truthful Camporesi–Higuchi spatial Dirac on $\mathcal{H}^{n_{\max}}_{\mathrm{GV}}$, diagonal in the spinor basis with eigenvalue $\chi(n+1/2)$ where $\chi \in \{+1,-1\}$ is the spinor chirality and n the Fock-shell index. Krein-self-adjointness $D_L^\times := J_L\, D_L^\dagger\, J_L = D_L$ is verified by direct computation: $\gamma^{0\,\dagger} = \gamma^0$ (Hermitian chirality-swap), $\{\gamma^0, D_{\mathrm{GV}}\} = 0$ in the chiral basis, $\partial_t^\dagger = -\partial_t$, and $J_L^2 = +I$ combine to give the stated identity. The sign $i^t = +i$ in (5) is forced by this requirement (an opposite sign would give an anti-Krein-self-adjoint D_L); see Paper 43 § 3 [41] for the explicit derivation.

2.4 Truncated Lorentzian operator system

Per Paper 44 [42], the truncated Lorentzian operator system is the pure-tensor span

$$\mathcal{O}^L_{n_{\max},N_t,T} \;:=\; \mathrm{span}_{\mathbb{C}}\big\{M^{\mathrm{spat}}_{N,L,M} \otimes M^{\mathrm{temp}}_p \;:\; (N,L,M) \in \mathcal{I}^{\mathrm{spat}}_{n_{\max}},\; p \in \{0,1,\ldots,N_t-1\}\big\}, \tag{7}$$

where each spatial multiplier $M^{\rm spat}_{N,L,M} = \mathrm{blkdiag}(W_{N,L,M}, W_{N,L,M})$ is the chirality-doubled lift of the Avery–Wen–Avery Weyl-sector multiplier $W_{N,L,M}$ (the Clebsch–Gordan / Avery 3-Y integral construction of Paper 32 [36] § III), and the temporal multipliers $M^{\rm temp}_p$ are momentum-polynomial diagonal:

$$M^{\rm temp}_p \;=\; \mathrm{diag}\big(\omega_k^p\big)_{|k|\le K_{\max}}. \tag{8}$$

Vandermonde invertibility at N_t distinct frequencies ω_k ensures that the span (7) coincides with the span of Fourier-mode indicators $M^{\rm temp}_q = \mathrm{diag}(\delta_{k,q})$, and we use whichever basis is more convenient.

By Paper 44 § 3, $\mathcal{O}^L_{n_{\max},N_t,T}$ is a $*$-closed linear subspace of $\mathcal{B}(\mathcal{K}_{n_{\max},N_t})$, contains the identity, and is *not* multiplicatively closed (a witness pair is constructed in Paper 44 § 5). The propagation number is $\mathrm{prop}(\mathcal{O}^L_{n_{\max},N_t}) = 2$ on the Weyl-doubled achievable envelope, matching the Riemannian Paper 32 § III prop=2 verbatim; prop $= \infty$ on the full $\mathcal{B}(\mathcal{K})$ envelope due to the chirality-doubling and the commutative temporal subalgebra.

2.5 Joint central spectral Fejér kernel

The SU(2) factor is the Paper 38 [37] L2-DEF central spectral Fejér kernel

$$K^{\rm SU(2)}_{n_{\max}}(g) \;:=\; \frac{1}{Z^{\rm SU(2)}_{n_{\max}}}\Big|\sum_{j\le j_{\max}} \sqrt{2j+1}\,\chi_j(g)\Big|^2, \tag{9}$$

with $j_{\max} = (n_{\max}-1)/2$ and $Z^{\rm SU(2)}_{n_{\max}} = n_{\max}(n_{\max}+1)/2$. Its *Plancherel mass distribution* (the squared Plancherel weights of the amplitude $h = \sum_{j\le j_{\max}} \sqrt{2j+1}\,\chi_j$, a probability distribution on the truncated dual) is

$$m^{\rm SU(2)}_{n_{\max}}(j) \;=\; \frac{2j+1}{Z^{\rm SU(2)}_{n_{\max}}}\,\mathbf{1}_{\{j\le j_{\max}\}}. \tag{10}$$

The multiplier (Fourier) symbol $\sigma^{\rm SU(2)}_{n_{\max}}(j')$ of $K^{\rm SU(2)}$ — the scalar by which convolution against $K^{\rm SU(2)}$ acts on the j'-isotypic block — is a *different* object: it is given by the Clebsch–Gordan sum recorded in Appendix A, satisfies $\sigma^{\rm SU(2)}(0) = 1$ (forced by normalization), $0 \le \sigma^{\rm SU(2)} \le 1$, and is supported on $j' \le 2j_{\max}$. At $n_{\max} = 2$ the values are $(1, \sqrt{2}/3, 2/9)$ at $j' = (0, \frac{1}{2}, 1)$. The two objects must not be conflated (Lemma 4.1; § 5.3).

The $U(1)$ factor is the standard Fejér kernel on S^1_T,

$$K^{U(1)}_{N_t,T}(\theta) \;:=\; \frac{1}{N_t}\Big|\sum_{|k|\le K_{\max}} e^{i\,2\pi k\theta/T}\Big|^2, \qquad K_{\max} = (N_t-1)/2 \;\;(N_t \text{ odd}), \tag{11}$$

with multiplier symbol

$$\sigma^{U(1)}_{N_t}(k) \;=\; \max\!\Big(0,\, \tfrac{N_t-|k|}{N_t}\Big) \tag{12}$$

and uniform Plancherel mass distribution $m^{U(1)}_{N_t}(k) = N_t^{-1}\mathbf{1}_{\{|k|\le K_{\max}\}}$. The joint kernel is the tensor product

$$K^{\rm joint}_{n_{\max},N_t,T}(g,\theta) \;:=\; K^{\rm SU(2)}_{n_{\max}}(g)\cdot K^{U(1)}_{N_t,T}(\theta), \tag{13}$$

with factorized multiplier symbol

$$\sigma^{K^{\rm joint}}(j',k) \;=\; \sigma^{\rm SU(2)}_{n_{\max}}(j')\cdot\sigma^{U(1)}_{N_t}(k), \tag{14}$$

and factorized mass distribution $m^{K^{\rm joint}} = m^{\rm SU(2)}\cdot m^{U(1)}$. Both factors are non-negative, normalized (L^1 norm $= 1$), and central (class functions on their respective groups); the tensor product inherits these properties on $\mathrm{SU}(2)\times U(1)$. The two conventions must not be mixed: pairing the SU(2) *mass* maximum ($2/(n_{\max}+1)$ at $j_{\max}$) with the $U(1)$ *symbol* maximum (1 at $k = 0$) produces a hybrid quantity that is neither a mass maximum (that is $2/(n_{\max}+1)\cdot N_t^{-1}$) nor a cb-norm (that is 1; Lemma 4.1).

2.6 Krein-positive subspace and weak-form propinquity

The Krein-positive subspace is

$$\mathcal{K}^+ \;:=\; \{|\psi\rangle \in \mathcal{K} \;:\; J_L|\psi\rangle = +|\psi\rangle\}, \tag{15}$$

the $+1$-eigenspace of J_L, of dimension $\dim \mathcal{K}^+ = \dim \mathcal{K}/2 = \frac{1}{3}\, n_{\max}(n_{\max}+1)(n_{\max}+2)N_t$. The Krein inner product $\langle \psi, \phi\rangle_{J_L} := \langle \psi, J_L \phi\rangle$ restricted to $\mathcal{K}^+$ reduces to the standard Hilbert inner product $\langle \psi, \phi\rangle$, since $J_L|\phi\rangle = |\phi\rangle$ on $\mathcal{K}^+$. Hence $\mathcal{K}^+$ is a genuine Hilbert space.

The orthogonal projection onto $\mathcal{K}^+$ is $P_+ := (I + J_L)/2$ (and dually $P_- := (I - J_L)/2$ for the -1-eigenspace).

Definition 2.1 ($\mathcal{K}^+$-restricted weak-form Lorentzian propinquity)**.** For two Lorentzian truncated Krein metric spectral triples $\mathcal{T}_i^L = (\mathcal{O}_i^L, \mathcal{K}_i, D_i^L, J_i^L)$ with the construction above, the $\mathcal{K}^+$-restricted weak-form Lorentzian propinquity is

$$\Lambda^L\big(\mathcal{T}_1^L, \mathcal{T}_2^L\big) \;:=\; \Lambda\big(\mathcal{T}_1^+, \mathcal{T}_2^+\big), \tag{16}$$

where $\mathcal{T}_i^+ := (P_i^+ \mathcal{O}_i^L P_i^+, \mathcal{K}_i^+, P_i^+ D_i^L P_i^+)$ is the $\mathcal{K}^+$-compressed candidate triple and Λ is a quantum Gromov–Hausdorff-type distance on metric spectral triples (the applicable Riemannian framework is van Suijlekom's state-space distance [33]; Latrémolière's propinquity [16] imposes tunnel hypotheses not available here).

Definition 2.1 is the natural candidate — and it is *degenerate*: Theorem 5.2 shows the compressed triple $\mathcal{T}^+$ has Lipschitz seminorm identically zero on its operator system, so it is not a compact quantum metric space in any framework (Latrémolière's or van Suijlekom's), and Λ^L as defined is not a distance. The operator-level J_L-equivariance that makes the compression well-defined (every Berezin-image operator and every truncation projection commutes with J_L; Paper 44 § 4) is genuine — but what it preserves carries no metric content.

2.7 Joint gradient norms

For a pure-tensor symbol $f = f_s \otimes f_t$ on $S^3 \times S^1_T$, the L^1-additive and L^2-Pythagorean joint gradient norms are

$$\|\nabla^{\text{joint},L^1} f\|_\infty := \|\nabla_x f_s\|_\infty \cdot \|f_t\|_\infty + \|f_s\|_\infty \cdot \|\partial_t f_t\|_\infty, \tag{17}$$

$$\|\nabla^{\text{joint},L^2} f\|_\infty := \sup_{(x,t)} \sqrt{|\nabla_x f(x,t)|^2 + |\partial_t f(x,t)|^2}. \tag{18}$$

For general f, both extend by linearity over the Peter–Weyl $\times$ Fourier expansion and obey the standard triangle inequalities. The L^2 form is dominated by the L^1 form via $\sqrt{a^2+b^2} \le a + b$ for $a, b \ge 0$; both forms upper-bound the same Lipschitz seminorm with the same constant in the proof of Lemma 4.4.

3 Lemma L1′: the operator-system substrate

Lemma L1′ supplies the truncated metric spectral triple on which the four subsequent lemmas operate. It is the closure result of Paper 44 [42], summarized here for completeness.

Lemma 3.1 (L1′, Paper 44)**.** *The collection $\mathcal{T}^L_{n_{\max},N_t,T} := (\mathcal{O}^L_{n_{\max},N_t,T}, \mathcal{K}_{n_{\max},N_t}, D^L_{n_{\max},N_t,T}, J^L_{n_{\max},N_t})$ defined in § 2.2–2.4 is a Lorentzian truncated Krein metric spectral triple in the sense of vdD [32] / BBB [3]:*

(a) $\mathcal{O}^L$ is a unital $*$-closed linear subspace of $\mathcal{B}(\mathcal{K})$.

(b) D_L *is Krein-self-adjoint,* $D_L^\times = D_L$*.*

(c) J_L *is a fundamental symmetry,* $J_L^2 = +I$*,* $J_L^\dagger = J_L$*.*

(d) The propagation number $\mathrm{prop}(\mathcal{O}^L) = 2$ *on the Weyl-doubled achievable envelope.*

(e) Every multiplier in $\mathcal{O}^L$ *commutes with* J_L*:* $[J_L, a] = 0$ *bit-exact for every* $a \in \mathcal{O}^L$*.*

(f) At $N_t = 1$*, the construction reduces bit-exactly to the Riemannian chirality-doubled truncated spectral triple on* $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ *(Frobenius residual* 0.0 *in float64).*

Proof. (a)–(c) are direct from the construction and Paper 43 § 2,3,5. (d) is Paper 44 § 3 (via the structural identity $\dim((\mathcal{O}^L)^2) = \dim^2_{\mathrm{Weyl}} \cdot N_t$ on the Weyl-doubled envelope). (e) follows from $[\gamma^0, M^{\mathrm{spat}}] = 0$ (chirality-swap acts trivially on the diagonal block-doubled spatial multiplier) and $[I_{N_t}, M^{\mathrm{temp}}] = 0$ (trivially). (f) is Paper 44 § 2 (load-bearing falsifier F1). $\square$

Remark 3.2 (Krein-positivity at the operator-multiplier level is trivial)**.** By Lemma 3.1(e), every operator $a \in \mathcal{O}^L$ preserves both $\mathcal{K}^+$ and $\mathcal{K}^-$. Consequently, the $\mathcal{K}^+$-restricted operator system $\mathcal{O}^{L,+} := P_+\mathcal{O}^L P_+$ has the same propagation number 2 as the unrestricted system. No quantum-metric machinery, however, applies to $(\mathcal{O}^{L,+}, \mathcal{K}^+, P_+D_LP_+)$: the compressed Dirac $P_+D_LP_+$ loses its entire spatial part (Theorem 5.2(i)), the induced Lipschitz seminorm vanishes identically, and no quantum-metric machinery — Latrémolière's or van Suijlekom's — has its hypotheses satisfied on this triple.

Remark 3.3 (Validity of the $\mathcal{K}^+$-restricted tunneling pair)**.** Every element of the standard Latrémolière tunneling pair $(B^{\mathrm{joint}}, P^{\mathrm{joint}})$ used in § 5 commutes with J_L at the operator level: $[J_L, B^{\mathrm{joint}}(f)] = 0$ for every $f \in C^\infty(S^3 \times S^1_T)$ (this is Lemma 4.10(e), the joint Berezin's Krein-positivity preservation, proved factor-wise via $[\gamma^0, M^{\mathrm{spat}}_{N,L,M}] = 0$ on the chirality-doubled spatial block and $[I_{N_t}, M_q^{\mathrm{temp}}] = 0$ on the temporal slot), and $[J_L, P^{\mathrm{joint}}] = 0$ because $P^{\mathrm{joint}} = P^{\mathrm{SU(2)}}_{n_{\max}} \otimes P^{U(1)}_{N_t}$ is a tensor of Stinespring orthogonal projections into the same multiplier-label subspaces preserved by B^{joint}. Consequently the $\mathcal{K}^+$-restricted pair $(P_+\, B^{\mathrm{joint}}(\cdot)\, P_+,\ P_+\, P^{\mathrm{joint}}(\cdot)\, P_+)$ is well-defined as a UCP pair on the compressed objects of Definition 2.1 — the trivial-multiplier $\mathcal{K}^+$-preservation feature inherited from the L1′ substrate construction of Paper 44 [42] § 3–4. The equivariance is genuine, but the structure it preserves carries a Lipschitz seminorm that is identically zero (Theorem 5.2), so there is no metric bookkeeping to transport. Equivariance without separation is the precise failure mode.

4 The three analytical lemmas

We now prove the three analytical lemmas L2 (joint cb-norm, § 4.1), L3 (joint Lichnerowicz, § 4.2), and L4 (joint Berezin reconstruction, § 4.3) on the substrate supplied by Lemma 3.1. The mechanisms transfer from Paper 38 / Paper 40 with one substantive simplification per lemma, arising from the pure-tensor structure of $\mathcal{O}^L$ and the momentum-diagonal structure of the temporal multipliers.

4.1 Lemma L2: joint cb-norm of the central Schur multiplier

Lemma 4.1 (L2: smoothing map and mass distribution)**.** *For all* $n_{\max} \geq 1$ *and odd* $N_t \geq 1$*:*

(a) The joint kernel K^{joint} *is non-negative, central, and normalized:* $\|K^{\mathrm{joint}}\|_{L^1} = 1$*.*

(b) The smoothing map $S_{K^{\mathrm{joint}}} : f \mapsto K^{\mathrm{joint}} * f$ *(equivalently, the central Fourier multiplier with symbol* $\sigma^{K^{\mathrm{joint}}}$ *of (14)) is unital and completely positive, hence*

$$\|S_{K^{\mathrm{joint}}_{n_{\max},N_t,T}}\|_{\mathrm{cb}} \;=\; 1. \tag{19}$$

Its multiplier symbol satisfies $\sigma^{K^{\text{joint}}}(0,0) = 1$*,* $0 \le \sigma^{K^{\text{joint}}} \le 1$*, with support* $j' \le 2j_{\max}$*,* $|k| \le N_t - 1$*.*

(c) The joint Plancherel mass distribution $m^{K^{\text{joint}}}(j,k) = m^{\mathrm{SU}(2)}(j)\, m^{U(1)}(k)$ *is a probability distribution on the truncated joint dual with maximum* $\frac{2}{n_{\max}+1} \cdot \frac{1}{N_t}$*, attained at* $(j,k) = (j_{\max}, \text{any } |k| \le K_{\max})$*.*

Proof. (a) Both factors are $|\cdot|^2/Z$-normalized; Haar measure on the product factorizes. (b) $S_{K^{\text{joint}}}$ is convolution against a probability density: positivity is immediate, and unitality ($S_{K^{\text{joint}}}\mathbf{1} = \mathbf{1}$, i.e. $\sigma^{K^{\text{joint}}}(0,0) = \int K^{\text{joint}} = 1$) is forced by (a). A unital positive map on a commutative C*-algebra is completely positive, and a unital completely positive map has cb-norm exactly 1 [24]. On the truncated noncommutative side, the induced central multiplier acts block-scalar on $\bigoplus_\pi M_{d_\pi}(\mathbb{C})$, and a block-scalar map with symbol σ has cb-norm $\sup_\pi |\sigma(\pi)| \le 1$, with equality attained on the trivial block. The symbol bounds and support are Appendix A. (c) Direct from (10) and $m^{U(1)} = N_t^{-1}$ uniform: the SU(2) mass $(2j+1)/Z$ increases in j with maximum $n_{\max}/Z^{\mathrm{SU}(2)}_{n_{\max}} = 2/(n_{\max}+1)$ at $j_{\max}$. □

Remark 4.2 (The mass maximum is not a cb-norm)**.** The mass-distribution maximum $2/(n_{\max}+1)$ of Lemma 4.1(c) is not the cb-norm of any map appearing in this paper; the smoothing map is unital completely positive with cb-norm 1 (Lemma 4.1(b)), an elementary statement requiring no amenability input. Conflating the two objects has two consequences worth flagging: (i) it would assign the UCP map B^{joint} a "cb-norm" below 1, which is impossible; (ii) it would suggest a bounded partial inverse of B^{joint} "at the same scale" — invalid, since an inverse is governed by the symbol *minimum* on the relevant window (see § 5.3).

Remark 4.3 ($4/\pi$ asymptote on the SU(2) factor)**.** Paper 38 § 3.2 (with the universal Plancherel-×-Vandermonde cancellation theorem from Paper 40 § 3.2 [39]) gives the asymptotic rate constant

$$\lim_{n_{\max}\to\infty} \frac{n_{\max} \cdot \gamma^{\mathrm{SU}(2)}_{n_{\max}}}{\log n_{\max}} = \frac{4}{\pi}.$$

This is the universal compact-Lie-group rate constant on the bi-invariant metric and is independent of rank, Weyl-group order, and Lie type within the class. The $U(1)$ factor inherits a separate Fejér-on-the-circle rate.

4.2 Lemma L3: joint Lichnerowicz / Lipschitz comparison

Lemma 4.4 (L3, joint Lichnerowicz)**.** *For every* $a \in \mathcal{O}^L_{n_{\max},N_t,T}$ *and every* $(n_{\max}, N_t, T)$*,*

$$\|[D_L, a]\|_{\text{op}} \;\le\; C_3^{\text{joint}}(n_{\max}, N_t) \cdot \|\nabla^{\text{joint}} a\|_\infty, \tag{20}$$

where ∇^{joint} *is either the* L^1*-additive or* L^2*-Pythagorean joint gradient norm (17), (18). The constant satisfies*

$$C_3^{\text{joint}}(n_{\max}, N_t) \;\le\; C_3^{\mathrm{SU}(2)}(n_{\max}) \;\le\; \sup_{2\le N\le 2n_{\max}-1} \frac{N-1}{\sqrt{N^2-1}} \;=\; \sqrt{1 - \frac{1}{n_{\max}}} \;\xrightarrow{n_{\max}\to\infty}\; 1^-, \tag{21}$$

where the supremum is taken over the natural-substrate envelope $N \le 2n_{\max} - 1$ *— the Avery–Wen–Avery* $\Delta n = \pm 1$ *ladder on* $n_{\max}$ *shells produces shell-difference labels* N *up to* $2n_{\max} - 1$ *via the chirality-doubling at the operator-system level (see Paper 44 [42] § 3). The per-harmonic function* $(N-1)/\sqrt{N^2-1}$ *is Paper 38 [37] Lemma L3 verbatim; the envelope range* $N \le 2n_{\max} - 1$ *extends Paper 38's* $N \le n_{\max}$ *because Paper 38's Berezin map restricts to multiplier labels* $N \le n_{\max}$ *by Plancherel cutoff, while the Lorentzian natural substrate's full multiplier algebra reaches the larger envelope (see Remark 4.5 below).*

Remark 4.5 (On the supremum range)**.** The supremum is taken over the natural-substrate envelope $N \le 2n_{\max} - 1$ rather than $N \le n_{\max}$. The per-harmonic functional form, the asymptotic limit 1^-, and the rate expressions of § 5.3 (eq. 30) are insensitive to this distinction. The finite-cutoff constant is sharpened by the envelope: at $n_{\max} = 3$, the natural-substrate generator $(N, L, M, p) = (4, 1, \pm 1, 0)$ has Lipschitz ratio 0.95, which exceeds the per-shell value $C_3^{(3)} = 2/\sqrt{8} = 0.707$ but is bounded by the envelope-aware $C_3^{\mathrm{op}}(3) = ((2\cdot 3-1)-1)/\sqrt{(2\cdot 3-1)^2 - 1} = 4/\sqrt{24} = \sqrt{2/3} \approx 0.816$. Numerical-panel values in §6.1 are unaffected (they quote γ^{joint} directly, not $C_3^{\mathrm{joint}} \cdot \gamma^{\mathrm{joint}}$). The envelope refinement was identified in Paper 46 [43] § 4.3 (the strong-form companion under $L_{\mathrm{op}}(a) := \|[D_L, a]\|_{\mathrm{op}}$); Paper 38 is unaffected because Paper 38's Berezin map restricts multiplier labels to $N \le n_{\max}$ by Plancherel cutoff, while the Lorentzian natural substrate's chirality-doubled multiplier algebra reaches the larger envelope (see Paper 46 Appendix A for the cross-paper analysis).

Proof. We first establish the structural identity: for every pure-tensor $a = a_s \otimes a_t \in \mathcal{O}^L$,

$$[D_L, a_s \otimes a_t] \;=\; i\,[D_{\mathrm{GV}}, a_s] \otimes a_t \qquad \text{(bit-exact).} \tag{22}$$

Expand $[D_L, a_s \otimes a_t] = i\,[\gamma^0 \otimes \partial_t, a_s \otimes a_t] + i\,[D_{\mathrm{GV}} \otimes I_{N_t}, a_s \otimes a_t]$ and analyze each term.

Term A (vanishing cross term). Since $a_s = \mathrm{blkdiag}(W, W)$ has identical diagonal blocks in chirality and γ^0 is the chirality-swap acting as $\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$ on the chirality indices,

$$\gamma^0\, a_s \;=\; \begin{pmatrix} 0 & W \\ W & 0 \end{pmatrix} \;=\; a_s\, \gamma^0, \qquad [\gamma^0, a_s] = 0.$$

Furthermore, a_t and ∂_t are both diagonal in the momentum basis (the temporal multiplier algebra is the commutative diagonal subalgebra of $M_{N_t}(\mathbb{C})$), so $[\partial_t, a_t] = 0$. The cross term therefore vanishes bit-exactly: $[\gamma^0 \otimes \partial_t, a_s \otimes a_t] = 0$.

Term B (purely spatial). $[D_{\mathrm{GV}} \otimes I_{N_t}, a_s \otimes a_t] = [D_{\mathrm{GV}}, a_s] \otimes a_t$.

Combining gives (22). By linearity over the operator-system span, the same identity extends to every $a \in \mathcal{O}^L$ written as a finite linear combination of pure-tensor generators.

Tensor-product operator-norm factorization $\|A \otimes B\|_{\mathrm{op}} = \|A\|_{\mathrm{op}} \cdot \|B\|_{\mathrm{op}}$ and Paper 38 Lemma L3 on the spatial factor give, for any pure-tensor $a = a_s \otimes a_t$,

$$\|[D_L, a]\|_{\mathrm{op}} \;=\; \|[D_{\mathrm{GV}}, a_s]\|_{\mathrm{op}} \cdot \|a_t\|_{\mathrm{op}} \;\le\; C_3^{\mathrm{SU}(2)}(n_{\max}) \cdot \|\nabla_x f_s\|_\infty \cdot \|a_t\|_{\mathrm{op}}, \tag{23}$$

where f_s is the Berezin pre-image of a_s. For momentum-diagonal a_t, $\|a_t\|_{\mathrm{op}} = \|f_t\|_\infty$ (with f_t the corresponding Fourier symbol). Since $\|\nabla_x f_s\|_\infty \cdot \|f_t\|_\infty \le \|\nabla^{\mathrm{joint}, L^1} f\|_\infty$ by (17) (the second summand is non-negative), and similarly $\le \|\nabla^{\mathrm{joint}, L^2} f\|_\infty$ by $\sqrt{a^2 + b^2} \ge a$ for $a, b \ge 0$, we obtain (20) for pure-tensor a. The extension to general $a = \sum_j c_j (a_s^j \otimes a_t^j)$ follows by triangle inequality on both sides.

The bound (21) takes the Paper 38 § L3 per-harmonic form $(N-1)/\sqrt{N^2 - 1}$ on the natural-substrate envelope $N \le 2n_{\max} - 1$ (per Remark 4.5); asymptotic tightness $\to 1^-$ is via the per-harmonic Avery–Wen–Avery saturation on $Y^{(3)}_{N,0,0}$ as $N \to 2n_{\max} - 1$. □

Remark 4.6 (The temporal direction contributes zero)**.** The identity (22) is the substantive structural content of Lemma 4.4. There is *no* cross term and *no* separate $U(1)$ Lichnerowicz lemma: the joint commutator is the spatial commutator times the temporal operator-norm, and the temporal slot enters the bound multiplicatively at value 1. This is stronger than the $C_3 \le 1 + \text{cross}$ estimate one would expect from a generic tensor extension; the structural reason is that the temporal multiplier algebra is commutative diagonal and ∂_t is itself diagonal, so they commute bit-exactly. The Wick-rotated convention here is essential: a Lorentzian-causal temporal derivative would not be diagonal in the same basis.

Remark 4.7 (L^1 and L^2 give the same C_3). Both L^1-additive (17) and L^2-Pythagorean (18) joint gradient norms upper-bound the right-hand side of (23) by the same constant $C_3^{\mathrm{SU}(2)}$: the temporal contribution to $\|[D_L, a]\|_{\mathrm{op}}$ is zero (by (22)), so only the spatial part of the joint gradient matters in the inequality.

4.3 Lemma L4: joint Berezin reconstruction

Definition 4.8 (Joint Berezin map, convolution form). The joint Berezin map is

$$B^{\mathrm{joint}}_{n_{\max},N_t,T}(f) \;:=\; P^{\mathrm{joint}}\, M_{K^{\mathrm{joint}}*f}\, P^{\mathrm{joint}}, \tag{24}$$

where M_g is multiplication by g on $L^2(S^3 \times S^1_T)$ and $P^{\mathrm{joint}} := P^{\mathrm{SU}(2)}_{n_{\max}} \otimes P^{U(1)}_{N_t}$ is the joint truncation projection onto the subspace spanned by the kept Peter–Weyl $\times$ Fourier modes.

Remark 4.9 (Sum-form pitfalls: weights and targets). Two natural-looking sum-form alternatives to (24) fail, and both failures are instructive. (i) *Weights:* a sum form weighted by the mass distribution $m^{\mathrm{SU}(2)}(N)$ in place of the multiplier symbol $\sigma^{\mathrm{SU}(2)}(N)$ is not unital ($B^{\mathrm{joint}}(\mathbf{1}) = Z^{-1}I \neq I$) and maximally violates the approximate-identity property at $f = \mathbf{1}$. (ii) *Targets:* under the convolution form, the image of the Fourier mode e^{iqt} is $\sigma^{U(1)}(q)$ times the *truncated shift* (Toeplitz) operator on $\mathbb{C}^{N_t}$ — not momentum-diagonal for $q \neq 0$ — so B^{joint}'s temporal image is *not contained* in the operator system $\mathcal{O}^L$ of § 2.4, whose temporal factor is momentum-diagonal; and a sum form assigning e^{iqt} a momentum-diagonal image is not the compression of any multiplication operator. This target mismatch is part of the degeneracy diagnosis of § 5: the momentum-diagonal operator system is not the algebra the Berezin map naturally reconstructs, and that choice is precisely what makes the temporal factor invisible to the Dirac commutator (Lemma 4.4). Rebuilding the construction with the Toeplitz temporal algebra is the reopened target of § 7.3.

Lemma 4.10 (L4, joint Berezin properties). $B^{\mathrm{joint}}_{n_{\max},N_t,T}$ *satisfies the following four properties.*

(a) Positivity. *If $f \geq 0$ pointwise on $S^3 \times S^1_T$, then $B^{\mathrm{joint}}(f) \geq 0$ as a positive-semidefinite Hermitian element of $M_{\dim \mathcal{K}}(\mathbb{C})$.*

(b) Contractivity. *B^{joint} is unital completely positive; $\|B^{\mathrm{joint}}\|_{\mathrm{cb}} = 1$ and $\|B^{\mathrm{joint}}(f)\|_{\mathrm{op}} \leq \|f\|_\infty$.*

(c) Approximate identity. *For every $f \in C^\infty(S^3 \times S^1_T)$,*

$$\|B^{\mathrm{joint}}(f) - P^{\mathrm{joint}}\, M_f\, P^{\mathrm{joint}}\|_{\mathrm{op}} \;\leq\; \gamma^{\mathrm{joint}}_{n_{\max},N_t,T} \cdot \|\nabla^{\mathrm{joint}} f\|_\infty, \tag{25}$$

with $\gamma^{\mathrm{joint}} = O(\log n_{\max}/n_{\max} + T/N_t) \to 0$ as $(n_{\max}, N_t) \to (\infty, \infty)$.

(d) Spatial L3 compatibility. *For every spatially-pure $f = f_s \otimes \mathbf{1}$,*

$$\|[D_L, B^{\mathrm{joint}}(f)]\|_{\mathrm{op}} \;\leq\; C_3^{\mathrm{joint}} \cdot \|\nabla^{\mathrm{joint}} f\|_\infty, \tag{26}$$

with the same $C_3^{\mathrm{joint}} \leq 1$ as in Lemma 4.4. (For temporally-varying f the convolution-form image is Toeplitz on the temporal factor, where $[\gamma^0 \otimes \partial_t, \cdot]$ does not vanish; the required estimate is precisely the Connes–van Suijlekom S^1-type Toeplitz commutator bound [6] — part of the reopened target, § 7.3.)

(e) Krein-positivity preservation. *For every f, $[J_L, B^{\mathrm{joint}}(f)] = 0$ bit-exact, hence $B^{\mathrm{joint}}(f)$ preserves both $\mathcal{K}^+$ and $\mathcal{K}^-$.*

Proof. *(a) Positivity.* The convolution form $B^{\text{joint}}(f) = P^{\text{joint}}\,(K^{\text{joint}} * f)\,P^{\text{joint}}$ writes $B^{\text{joint}}(f)$ as a UCP compression of a non-negative function (since both factor kernels are non-negative pointwise: $K^{\text{SU}(2)} = |\cdot|^2 \cdot Z^{-1} \geq 0$ and $K^{U(1)} = |\cdot|^2 \cdot N_t^{-1} \geq 0$). The convolution of two non-negative integrands against the positive joint Haar measure is non-negative, and the UCP compression of a non-negative multiplication operator is positive-semidefinite.

(b) Contractivity. The convolution form gives $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|K^{\text{joint}} * f\|_\infty \leq \|K^{\text{joint}}\|_{L^1} \cdot \|f\|_\infty$ by Young's inequality at the sup endpoint, and $\|K^{\text{joint}}\|_{L^1} = 1$ (both factors are normalized probability densities; Haar on a product factorizes). B^{joint} is the composition of the unital smoothing map $S_{K^{\text{joint}}}$ (UCP with cb-norm 1, Lemma 4.1(b)) with the unital compression $g \mapsto P^{\text{joint}} M_g P^{\text{joint}}$ (UCP); a composition of UCP maps is UCP, so $\|B^{\text{joint}}\|_{\text{cb}} = 1$ with unitality $B^{\text{joint}}(\mathbf{1}) = P^{\text{joint}} M_{\mathbf{1}} P^{\text{joint}} = I$ on the truncated space.

(c) Approximate identity. For a pure-tensor symbol $f = f_s \otimes f_t$, the convolution-deficit splits:

$$K^{\text{joint}} * f - f \;=\; (K^{\text{SU}(2)} * f_s - f_s) \otimes (K^{U(1)} * f_t) + f_s \otimes (K^{U(1)} * f_t - f_t).$$

By Paper 38 [37] Lemma L4(c) on the spatial factor, $\|K^{\text{SU}(2)} * f_s - f_s\|_\infty \leq \gamma^{\text{SU}(2)}_{n_{\max}} \cdot \|\nabla_x f_s\|_\infty$, with $\gamma^{\text{SU}(2)}_{n_{\max}} = O(\log n_{\max}/n_{\max})$ (Paper 38 § L2 quantitative rate, $4/\pi$ asymptote). By the standard Fejér-on-the-circle approximation rate (Katznelson [15] Ch. I), $\|K^{U(1)} * f_t - f_t\|_\infty \leq \gamma^{U(1)}_{N_t,T} \cdot \|\partial_t f_t\|_\infty$ with $\gamma^{U(1)} = O(T/N_t)$. Combining with the L^1-additive triangle decomposition and $\|K^{U(1)} * f_t\|_\infty \leq \|f_t\|_\infty$ (Young) gives

$$\|K^{\text{joint}} * f - f\|_\infty \;\leq\; \max(\gamma^{\text{SU}(2)}_{n_{\max}}, \gamma^{U(1)}_{N_t,T}) \cdot \|\nabla^{\text{joint},L^1} f\|_\infty.$$

We define $\gamma^{\text{joint}} := \max(\gamma^{\text{SU}(2)}, \gamma^{U(1)})$. Compression-contraction on L^2 does not increase the operator norm, so (25) follows. Linearity and the triangle inequality extend to general f.

(d) Spatial L3 compatibility. For $f = f_s \otimes \mathbf{1}$, the temporal convolution of a constant is the constant and the temporal compression of $M_{\mathbf{1}}$ is I_{N_t}, so $B^{\text{joint}}(f) = B^{\text{SU}(2)}_{\chi\text{-d}}(f_s) \otimes I_{N_t}$ with $B^{\text{SU}(2)}_{\chi\text{-d}}$ chirality-block-diagonal with equal blocks. Then $[\gamma^0 \otimes \partial_t, B^{\text{SU}(2)}_{\chi\text{-d}}(f_s) \otimes I_{N_t}] = [\gamma^0, B^{\text{SU}(2)}_{\chi\text{-d}}(f_s)] \otimes \partial_t = 0$, and the spatial term is bounded by Paper 38 § L3-compatibility with the per-harmonic constant $C_3^{(N)} = (N-1)/\sqrt{N^2-1}$, giving (26).

(e) Krein-positivity preservation. For the convolution form, $[J_L, B^{\text{joint}}(f)] = P^{\text{joint}}\,[J_L, M_{K^{\text{joint}} * f}]\,P^{\text{joint}}$ (since $[J_L, P^{\text{joint}}] = 0$), and $[J_L, M_g] = 0$ for every function g because $J_L = J_L^{\text{spat}} \otimes I_{N_t}$ acts on the chirality indices while M_g acts scalarly on them. □

Remark 4.11 (Tensor-product factorization on pure tensors)**.** For a pure-tensor $f = f_s \otimes f_t$, the joint Berezin factorizes:

$$B^{\text{joint}}(f_s \otimes f_t) \;=\; B^{\text{SU}(2)}_{\chi\text{-d}}(f_s) \otimes B^{U(1)}(f_t), \tag{27}$$

where $B^{\text{SU}(2)}_{\chi\text{-d}}$ is the chirality-doubled spinor lift of Paper 38's spatial Berezin and $B^{U(1)}$ is the Fejér–Toeplitz compression on S^1_T (the Connes–van Suijlekom circle object [6], *not* a momentum-diagonal map; cf. Remark 4.9). This factorization makes Lemma 4.10(a)–(c),(e) follow from the corresponding factor-wise properties of Paper 38 L4 (SU(2) side) and standard Fejér-on-S^1 approximation theory ($U(1)$ side); item (d) is spatial-only as stated.

5 The annihilation theorem and the failure of the compression assembly

5.1 The tunneling pair

Definition 5.1 (Joint tunneling pair)**.** The joint tunneling pair between the truncated triple $\mathcal{T}^L_{n_{\max},N_t,T}$ and the continuum reference triple $\mathcal{T}^L_{S^3\times S^1_T}$ is the pair $(B^{\text{joint}}_{n_{\max},N_t,T}, P^{\text{joint}}_{n_{\max},N_t})$, with

B^{joint} the joint Berezin map of Definition 4.8 and $P^{\text{joint}} = P^{\mathrm{SU}(2)}_{n_{\max}} \otimes P^{U(1)}_{N_t}$ the joint truncation projection on $L^2(S^3 \times S^1_T)$.

The continuum reference triple is $\mathcal{T}^L_{S^3\times S^1_T} := (C^\infty(S^3 \times S^1_T), L^2(S^3, \Sigma_{\mathrm{CH}}) \otimes L^2(S^1_T), D^L_\infty, J^L_\infty)$ with Σ_{CH} the chirality-doubled Camporesi–Higuchi spinor bundle and D^L_∞, J^L_∞ the natural continuum analogs of D_L and J_L.

By Lemmas 3.1–4.10, both B^{joint} and P^{joint} are UCP (cb-norm 1; the projection P^{joint} is a Stinespring orthogonal projection), and both commute with J_L at the operator level, so the $\mathcal{K}^+$-compression of the pair is well-defined. One might hope the compressed pair to be a tunneling pair between the compressed triples of Definition 2.1 in some Riemannian framework. It is not, twice over: no applicable framework admits a UCP-pair tunnel on these objects, and — more fundamentally — the compressed triples are not quantum metric spaces at all, as we now prove.

5.2 The annihilation theorem

Theorem 5.2 ($\mathcal{K}^+$ annihilation)**.** *Let $D_L = i(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t})$ with D_{GV} Hermitian, ∂_t anti-Hermitian and momentum-diagonal, and fundamental symmetry $J_L = J_L^{\text{spat}} \otimes I_{N_t}$, and suppose D_L is Krein-self-adjoint, $D_L^\times := J_L D_L^\dagger J_L = D_L$ (Lemma 3.1(b)). Then:*

- *(i) $\{J_L, D_{\mathrm{GV}} \otimes I_{N_t}\} = 0$, and consequently $P_+ (D_{\mathrm{GV}} \otimes I_{N_t}) P_+ = 0$: the $\mathcal{K}^+$ compression annihilates the spatial Dirac.*
- *(ii) The surviving compressed Dirac $P_+D_LP_+ = i\, P_+(\gamma^0 \otimes \partial_t)P_+$ is momentum-diagonal, and commutes with P_+aP_+ for every $a = M^{\text{spat}} \otimes M^{\text{temp}} \in \mathcal{O}^L$ (momentum-diagonal temporal factor).*
- *(iii) Therefore the $\mathcal{K}^+$-restricted Lipschitz seminorm vanishes identically on the operator system:*
$$\|[P_+D_LP_+,\ P_+aP_+]\|_{\mathrm{op}} \ = \ 0 \qquad \text{for every } a \in \mathcal{O}^L_{n_{\max},N_t,T}, \tag{28}$$
and the compressed pair $(P_+\mathcal{O}^LP_+,\ \|[P_+D_LP_+,\cdot]\|_{\mathrm{op}})$ is not a compact quantum metric space: the seminorm kernel is the whole system, the associated Monge–Kantorovich extended metric is degenerate, and Definition 2.1 does not define a quantum Gromov–Hausdorff-type distance.

Proof. (i) Write $D_s := D_{\mathrm{GV}} \otimes I_{N_t}$ (Hermitian) and $D_t := \gamma^0 \otimes \partial_t$. Since ∂_t is anti-Hermitian and $[J_L, D_t] = 0$ (both factors commute), the term iD_t is Hermitian and Krein-self-adjoint on its own. For the spatial term, $(iD_s)^\dagger = -iD_s$, so Krein-self-adjointness of D_L forces $J_L(-iD_s)J_L = iD_s$, i.e. $J_LD_sJ_L = -D_s$, i.e. $\{J_L, D_s\} = 0$. Then with $P_+ = (I + J_L)/2$,

$$P_+D_sP_+ \ = \ \tfrac{1}{4}\,(D_s + D_sJ_L + J_LD_s + J_LD_sJ_L) \ = \ \tfrac{1}{4}\,(D_s + 0 - D_s) \ = \ 0,$$

using $D_sJ_L + J_LD_s = 0$ and $J_LD_sJ_L = -D_s$. (ii) P_+ commutes with D_t and with every $a \in \mathcal{O}^L$ (Lemma 3.1(e)), so $P_+D_LP_+ = iD_tP_+$ restricted to $\mathcal{K}^+$; on $\mathcal{K}^+$, $J_L = +1$, so this operator is (unitarily) ∂_t-diagonal in the momentum basis. The temporal factors M^{temp} of $\mathcal{O}^L$ are momentum-diagonal by construction (§ 2.4), hence commute with it; the spatial factors commute with anything supported purely on the temporal slot. (iii) Immediate from (i) + (ii): the spatial part of the commutator is annihilated by (i), the temporal part vanishes by (ii). A seminorm whose kernel strictly exceeds $\mathbb{C}1$ fails the Rieffel/Latrémolière Lip-norm axiom ($L(a) = 0 \Leftrightarrow a \in \mathbb{C}1$) and the van Suijlekom state-space framework's separation requirement alike. □

Corollary 5.3 (Numerical verification, bit-exact)**.** *At the panel cells $(n_{\max}, N_t) \in \{(2, 3), (3, 5)\}$ the restricted seminorm (28) is 0.0 in `float64` for all 42/42 and 275/275 operator-system multipliers respectively, with $\|\{J_L, D_{\mathrm{GV}} \otimes I\}\|_F = 0.0$ and $\|P_+(i\,D_{\mathrm{GV}} \otimes I)P_+\|_F = 0.0$ exactly (uncompressed norms 15.87 and 43.47). The* unrestricted *seminorm $\|[D_L, \cdot]\|_{\mathrm{op}}$ is non-degenerate* *as a map but has kernel* 30/42 *and* 130/275 *— and already* 10/14*,* 26/55 *at* $N_t = 1$ *with the truthful chirality-diagonal Camporesi–Higuchi spatial Dirac — so the kernel condition fails before restriction as well. Driver:* `debug/p45_kplus_seminorm_check.py`*; frozen regression test:* `tests/test_p45_kplus_degeneracy.py`*.*

5.3 Anatomy of the failed assembly

We record where each piece of the natural assembly stands — both because the surviving estimates are genuine analytic facts about the kernels, and because the failure points specify the repair (§ 7.3). The assembly one would attempt is the bound

$$\Lambda\big(\mathcal{T}^+_{n_{\max},N_t,T}, \mathcal{T}^+_{S^3\times S^1_T}\big) \;\leq\; \max\big(\mathrm{reach}_B,\, \mathrm{reach}_P,\, \mathrm{height}_B,\, \mathrm{height}_P\big) \tag{29}$$

for the UCP pair $(B^{\mathrm{joint}}, P^{\mathrm{joint}})$, with the four constituents measured against the *classical* joint-gradient unit ball (an auxiliary third seminorm — not the Lipschitz ball of either triple, itself a structural defect of any such bookkeeping). No four-constituent UCP-pair tunnel exists in Latrémolière's framework [16] (tunnels there are carried by quantum metric spaces with quantum isometries); the nearest genuine framework is van Suijlekom's state-space Gromov–Hausdorff bound via approximation maps in both directions [18, 33]. Independently of the framework question, Theorem 5.2 shows the left-hand side of (29) is not a distance, so no assembly through Definition 2.1 can succeed. Piece by piece:

- *reach*$_B$ *(approximate-identity deficit).* The estimate $\|B^{\mathrm{joint}}(f) - P^{\mathrm{joint}} M_f P^{\mathrm{joint}}\|_{\mathrm{op}} \leq \gamma^{\mathrm{joint}} \|\nabla^{\mathrm{joint}} f\|_\infty$ (Lemma 4.10(c)) survives as a genuine approximation statement for the convolution-form Berezin map. It is an analytic fact about kernels, not a metric statement.
- *reach*$_P$ *(dual roundtrip).* No bounded partial inverse of B^{joint} can be inferred from any forward bound: an inverse is governed by the multiplier-symbol *minimum* on the relevant window (at full support the inverse norm grows like $O(n^2_{\max})$). The correct dual-direction mechanism is the antiderivative transference of Leimbach–van Suijlekom [18] (Lemmas 3.4–3.5 there); for scalar sectors on compact metric groups the conclusion is covered by Gaudillot-Estrada–van Suijlekom [10]. Adapting it to the chirality-doubled spinor substrate is a named gap shared with the companion S^3 paper.
- *height*$_B$ *(Lipschitz distortion).* Fails: for purely temporal $f = \mathbf{1} \otimes f_t$ with $\|\partial_t f_t\|_\infty = 1$, the classical side gives $\|f\|_{\mathrm{Lip}} = 1$ while $\|[D_L, B^{\mathrm{joint}}(f)]\|_{\mathrm{op}}$ vanishes on the momentum-diagonal reading, so the distortion is ≥ 1 and cannot tend to 0. The spatial restriction survives via Lemma 4.10(d).
- *height*$_P$ $= 0$. Vacuous under Theorem 5.2: every map into a zero-seminorm system is "Lipschitz-isometric."

The assembly therefore produces no distance bound. The rate expression it would have produced,

$$C_3^{\mathrm{joint}}(n_{\max}, N_t)\cdot\gamma^{\mathrm{joint}}_{n_{\max},N_t,T} \;=\; O\Big(\tfrac{\log n_{\max}}{n_{\max}} + \tfrac{T}{N_t}\Big) \;\xrightarrow[(n_{\max},N_t)\to(\infty,\infty)]{}\; 0, \tag{30}$$

is nonetheless a well-defined closed-form expression; its surviving meaning is the spatial convergence statement of Proposition 5.4, to which we now turn.

5.4 The spatial convergence statement

Proposition 5.4 (Spatial sector)**.** *Fix* $N_t \geq 1$*,* $T > 0$*. By the companion* S^3 *paper's (unconditional) spatial Gromov–Hausdorff convergence theorem (van Suijlekom state-space distance [33],* *with the truncated state space metrized by the left-translation Lipschitz seminorm; compression/lifted-state argument in the pattern of [10]), the spatial truncations underlying $\mathcal{T}^L_{n_{\max},N_t,T}$ converge to the spatial continuum reference at the Paper 38 rate*

$$\mathrm{d}^{\mathrm{vS}}_{\mathrm{GH}} \;\le\; C_3^{\mathrm{SU}(2)}(n_{\max})\cdot\gamma^{\mathrm{SU}(2)}_{n_{\max}} \;=\; O\Big(\tfrac{\log n_{\max}}{n_{\max}}\Big), \tag{31}$$

and the temporal factor is carried J_L-equivariantly (Lemma 4.10(e)) but is metrically invisible: it contributes nothing to any Dirac-commutator seminorm on the momentum-diagonal operator system (Lemma 4.4 structural identity; Theorem 5.2). In particular, no statement of Lorentzian or product-metric content is made: (31) is a Riemannian statement about the S^3 factor, decorated with Krein bookkeeping.

Proof sketch. At fixed N_t, the spatially-pure sector $f = f_s \otimes \mathbf{1}$ reduces, by the factorization of Remark 4.11 and Lemma 4.10(d), to Paper 38's spatial construction tensored with I_{N_t}; all estimates are the Paper 38 estimates verbatim, now unconditional under that paper's translation-seminorm metrization. The metric invisibility of the temporal factor is Theorem 5.2(ii)–(iii) together with the structural identity (22). □

Remark 5.5 (Asymptotic rate)**.** The joint rate factors at leading order:

$$\gamma^{\mathrm{joint}} \;\le\; \gamma^{\mathrm{SU}(2)} + \gamma^{U(1)} \;=\; O\Big(\tfrac{\log n_{\max}}{n_{\max}}\Big) + O\Big(\tfrac{T}{N_t}\Big),$$

with the SU(2) leading constant $\lim_{n_{\max}\to\infty} n_{\max}\gamma^{\mathrm{SU}(2)}_{n_{\max}}/\log n_{\max} = 4/\pi$ rigorous (Paper 38 § L2 Stein–Weiss closure + Paper 40 § 3.2 universal cancellation), and the $U(1)$ leading constant $\gamma^{U(1)}_{N_t,T} \le C\,T\log N_t/N_t$ standard (Katznelson [15] Ch. I).

Remark 5.6 (The compressed continuum object is also degenerate)**.** There is no limit identification to state: the candidate limit object $\mathcal{T}^+_{S^3\times S^1_T} = (P_+C^\infty(S^3\times S^1_T)P_+, \mathcal{K}^+_\infty, P_+D^L_\infty P_+)$ is itself degenerate — the compressed continuum Dirac loses its spatial part by the same anti-commutation mechanism as Theorem 5.2(i), so its Lipschitz seminorm kernel contains all time-independent functions and its Monge–Kantorovich structure does not separate states.

Remark 5.7 (Assembly layers are not mere bookkeeping)**.** In proof shapes of this kind it is tempting to regard the assembly layer as bookkeeping, with the analytical content concentrated in the kernel-level lemmas. The present construction inverts that assessment: the kernel-level facts (Lemma 4.10(a)–(c),(e)) are robust, while the assembly layer is where the construction fails — a degenerate seminorm, no admissible tunnel device, and a false constituent bound. Assembly steps that import external theorems are exactly the steps that require line-level hypothesis checking.

Remark 5.8 (Two-rate hybrid: what survives)**.** A two-rate hybrid composite to the non-compact carrier $S^3\times\mathbb{R}_t$ has two arrows. The outer (norm-resolvent) arrow of Paper 47 [44] § 4 is operator-level, does not use the Lipschitz seminorm, and is unaffected by the degeneracy, with rate $O(e^{-|\mathrm{Im}\,z|\,T/2})$ on compact-temporal-support subspaces. The inner (metric) arrow would require a Q1-repaired construction; the metric-level statement on both the compact and non-compact carriers is open (Paper 47 carries its own Status note).

5.5 Product-carrier convergence in the action-seminorm framework

The degeneracy theorem leaves a definite repair specification (§ 7.3, Q1(i)): a temporal multiplier algebra built from Toeplitz compressions of e^{iqt} rather than momentum-diagonal polynomials. That prerequisite has now been established, and with it the convergence content of the withdrawn assembly is restored at the *product-carrier* level — in the translation (action) seminorm framework of the companion S^3 paper's unconditional theorem, not in the Krein–Dirac-commutator framework that produced the degeneracy.

Proposition 5.9 (Product carrier, action seminorm)**.** *Metrize the truncated joint system $P\,(C(S^3\times S^1_T))\,P$, $P = P_{n_{\max}} \otimes P_K$, by the translation Lipschitz seminorm over the product isometry group $SU(2)\times U(1)$,*

$$L_{\mathrm{joint}}(A) \;=\; \sup_{|X|=1} \big\|[\,G(X),\,A\,]\big\|_{\mathrm{op}}, \qquad X \in \mathfrak{su}(2)\oplus\mathfrak{u}(1),$$

which equals the metric Lipschitz constant on the continuum. Then: (i) the kernel condition $L_{\mathrm{joint}}(A)=0 \Leftrightarrow A\in\mathbb{C}1$ holds on the truthful band-limited substrate ($S^3\times S^1$ is parallelizable, so the joint Killing fields frame the tangent space, and joint per-band injectivity is the product of the factor injectivities); (ii) the forward arrow is the plain compression, which commutes with the joint unitaries (the spatial spectral projections are $SU(2)$-equivariant, the Fourier window is translation-equivariant) and is therefore tautologically Lipschitz-contractive; (iii) the dual arrow is the product lifted state $\xi_s\otimes\xi_t$ (spatial spinor lifted state of the companion paper tensored with the temporal Fejér amplitude), whose almost-inverse defects are the two factor Fejér smoothings. Consequently the truncations converge to $C(S^3\times S^1_T)$ in van Suijlekom state-space Gromov–Hausdorff distance at the additive rate

$$\mathrm{d}_{\mathrm{GH}}^{\mathrm{vS}} \;\le\; \gamma^{\mathrm{SU}(2)}_{n_{\max}} + \gamma^{U(1)}_{N_t,T} \;=\; O\Big(\tfrac{\log n_{\max}}{n_{\max}}\Big) + O\Big(\tfrac{T\log N_t}{N_t}\Big) \;\longrightarrow\; 0, \tag{32}$$

in the pattern of [10] on each factor, with the $4/\pi$ spatial leading constant of Remark 5.5 unchanged.

Proof sketch and verification. Each ingredient is the tensor product of an unconditional factor ingredient: the spatial factor is the companion paper's theorem verbatim; the temporal factor is the Toeplitz algebra of Q1(i), for which the translation seminorm equals the continuum Lipschitz constant *exactly* on single modes, $L(S_q)=2\pi q/T$, at every finite window (the Connes–van Suijlekom S^1 pattern [6]; driver `debug/wh7_toeplitz_temporal_probe.py`, max error 1.4×10^{-14}). The joint claims are verified numerically on the $j\le 1$ Peter–Weyl window tensored with a $K=2$ Fourier window (`debug/wh7_b1_joint_product_gh.py`; frozen falsifier `tests/test_wh7_b1_joint.py`): Peter–Weyl orthonormality of the quadrature at 10^{-15}; pure-factor exactness of L_{joint} at 0 and 10^{-16}; the Leibniz envelope $\max(L_s\|B\|,\|A\|L_t)\le L_{\mathrm{joint}}(A\otimes B)\le L_s\|B\|+\|A\|L_t$ on a six-cell panel; the joint kernel condition; the additive smoothing bound $\|\Phi_s\otimes\Phi_t(F)-F\|\le(\gamma_s+\gamma_t)\,\mathrm{Lip}(F)$; and bit-exact reduction to the spatial statement at $N_t=1$ (the load-bearing Riemannian-limit falsifier of § 6.2, preserved in the rebuilt architecture). □

Remark 5.10 (What this restores, and what it does not)**.** Proposition 5.9 restores the convergence content the withdrawn assembly claimed — the temporal direction now carries genuine metric weight, because the seminorm is generated by the group action, which the Toeplitz multipliers do not commute with ($[D_t,S_q]=\omega_q S_q$, against $[D_t,g(D_t)]=0$ identically for the momentum-diagonal algebra). It does *not* make the statement Lorentzian: the translation seminorm is signature-blind, and the carrier is the Wick-rotated (thermal) product. The Krein-compression clause Q1(ii) is bypassed rather than solved — the action seminorm never routes the metric through the Dirac operator. The genuinely Lorentzian question is thereby sharpened (see the updated Q1): the signature must enter through the *generator* of the metric, and the natural candidate is the wedge boost — the modular flow of the four-witness theorem (Paper 42), already bit-exact at finite cutoff — i.e. a translation-type seminorm along modular orbits rather than parameter time.

$(n_{\max}, N_t)$	$\gamma^{\mathrm{SU}(2)}$	$\gamma^{U(1)}$	C_3^{joint}	cb-norm	Λ^L bound	$N_t = 1$ residual
$(2, 1)$	2.0746	$\pi/2 \approx 1.5708$	0.5774	2/3	2.0746	**0.0** bit-exact
$(3, 1)$	1.6101	$\pi/2$	0.7071	1/2	1.6101	**0.0** bit-exact
$(4, 1)$	1.3223	$\pi/2$	0.7746	2/5	1.3223	**0.0** bit-exact
$(2, 3)$	2.0746	0.7220	0.5774	2/3	2.0746	—
$(3, 5)$	1.6101	0.4956	0.7071	1/2	1.6101	—
$(4, 7)$	1.3223	0.3841	0.7746	2/5	1.3223	—

Table 1: Rate-formula panel. $\gamma^{\mathrm{SU}(2)}$ values from Paper 38 § L2 closed-form Stein–Weiss computation; $\gamma^{U(1)}$ values from standard Fejér-on-S^1_T at $T = 2\pi$; C_3^{joint} from the Avery harmonic form $(N-1)/\sqrt{N^2-1}$. The column labeled "cb-norm" tabulates $2/(n_{\max}+1)$, which is the SU(2) Plancherel *mass-distribution maximum*, not a cb-norm (the cb-norm of the smoothing map is 1 at every cell; Lemma 4.1). The "Λ^L bound" column is the closed-form expression $C_3^{\mathrm{joint}} \cdot \gamma^{\mathrm{joint}}$, whose surviving meaning is the spatial rate of Proposition 5.4; it is not an evaluated metric. The $N_t = 1$ residuals record the algebraic reduction of the formulas to Paper 38's formulas.

6 Numerical verification and Riemannian-limit recovery

6.1 Verification panel

The driver `debug/l3b_2_sub_sprint_D_compute.py` (project repository) evaluates the panel

$$(n_{\max}, N_t) \in \{(2,1), (3,1), (4,1), (2,3), (3,5), (4,7)\}$$

at $T = 2\pi$. The tabulated "Λ^L bound" values are *evaluations of the closed-form rate expressions* $\gamma^{\mathrm{SU}(2)}(n_{\max})$ and $\gamma^{U(1)}(N_t, T)$ (the production module `geovac/lorentzian_propinquity_compact_temporal.p` computes $C_3^{\mathrm{joint}} \cdot \gamma^{\mathrm{joint}}$ from kernel-moment formulas; no operator-level metric quantity is ever evaluated). The panel therefore verifies: (i) the algebraic reduction of the formulas at $N_t = 1$ to Paper 38's formulas; (ii) monotone decrease of the rate expression in $n_{\max}$; (iii) ratio consistency ≈ 0.637 with Papers 38/39; (iv) the assembly constituent inequalities of § 5.3 against the auxiliary classical-gradient ball. None of (i)–(iv) tests a metric statement; the operator-level content of the construction is tested instead by the annihilation falsifier `debug/p45_kplus_seminorm_check.py` (Corollary 5.3).

The numerical results are tabulated in Table 1 (data and driver in the project repository, `debug/data/l3b_2_sub_sprint_D.json`; see Appendix B for full data).

6.2 Riemannian-limit recovery (load-bearing falsifier)

Proposition 6.1 (Riemannian-limit reduction of the rate formulas). *At $N_t = 1$, the construction and its rate expression reduce bit-exactly to Paper 38's:*

$$C_3^{\mathrm{joint}}(n_{\max}, 1)\, \gamma^{\mathrm{joint}}_{n_{\max},1,T} \;=\; C_3^{\mathrm{SU}(2)}(n_{\max})\, \gamma^{\mathrm{SU}(2)}_{n_{\max}} \qquad \textit{(Frobenius residual } 0.0 \textit{ in float64).} \tag{33}$$

(Neither side is an evaluated metric — see § 6.1.)

Proof. At $N_t = 1$, the only kept Fourier mode is $q = 0$, the $U(1)$ Fejér kernel collapses to the uniform Haar density $1/T$, and the temporal Berezin factor $B^{U(1)}$ is the trivial 1×1 identity. By the tensor-product factorization (27), $B^{\mathrm{joint}}_{n_{\max},1,T}(f_s \otimes \mathbf{1}) = B^{\mathrm{SU}(2)}_{\chi\text{-d}}(f_s) \otimes I_1$, which equals Paper 38's chirality-doubled spinor lift of the spatial Berezin map.

The chirality-doubled spinor lift on $\mathcal{K}^+$ is bit-identical to Paper 38's scalar Fock-basis construction up to the basis representation: the kernel data ($m^{\mathrm{SU}(2)}$ and $\sigma^{\mathrm{SU}(2)}$), the Lipschitz bound constant $C_3^{\text{Paper 38}} \leq 1$, and the convergence rate $\gamma_{n_{\max}}^{\mathrm{SU}(2)}$ are the same numerical objects. Therefore (33) holds. The bit-exact residual at $n_{\max} \in \{2, 3, 4\}$ is verified numerically in Table 1. $\square$

On falsifier strength. Because the panel quantity only ever contains the spatial rate formula (§ 6.1), this $N_t = 1$ reduction is an automatic formula identity — it is *insensitive* to the degeneracy of the metric construction, which it cannot detect. A falsifier with actual teeth for this construction is the operator-level seminorm check of Corollary 5.3, which detects the failure immediately. The corresponding $N_t = 1$ reductions in Paper 42 (Tomita–Takesaki construction) and Paper 43 (Lorentzian Dirac) are genuinely operator-level.

6.3 Asymptotic rate consistency

The rate-formula ratio $1.3223/2.0746 = 0.6374$ is monotone decreasing across the panel and matches the single-factor Paper 38 and Paper 39 [38] ratios bit-identically — *because the tabulated quantity is the spatial formula*: the temporal factor never enters the maximum at these cells, and (by Theorem 5.2) never enters the operator-level seminorm at all.

7 Discussion

7.1 Four-witness Wick-rotation connection

Paper 42 [40] closes the four-witness Wick-rotation theorem (Hartle–Hawking [12], Sewell [27], Bisognano–Wichmann [2], Unruh [31]) at the operator-system level on the *Riemannian* truncated Camporesi–Higuchi spectral triple, via the geometric BW-α and Tomita–Takesaki BW-γ modular Hamiltonians on a hemispheric wedge. Paper 43 [41] extends that closure to signature $(3, 1)$ on the Krein wedge.

Papers 42/43 close modular-structure statements at the operator level; those are unaffected by the degeneracy. The convergence leg of the arc, by contrast, is *not closed*: the present paper proves that the natural device for it cannot work (Theorem 5.2) and states the open problem (§ 7.3).

7.2 Relation to Mondino–Sämann

The synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämann [20] extends their earlier work [19] to Lorentzian pre-length spaces via ϵ-nets of causal diamonds. This is adjacent at the conceptual level (both "Lorentzian Gromov–Hausdorff-type convergence") but operates on a categorically different mathematical object.

Remark 7.1 (Different objects, different distances)**.** Mondino–Sämann work on pre-length spaces with causal-diamond geometry; they do not have a Dirac operator or any spectral data, and their convergence is in the synthetic-geometry metric on the space of isometry classes of such pre-length spaces. The present paper works on truncated Krein spectral triples — and proves a *negative* structural result about the obvious operator-algebraic convergence device on them, rather than a convergence theorem. The two programs remain categorically disjoint; as of this writing the operator-algebraic side has *no* valid Lorentzian-type quantum Gromov–Hausdorff convergence notion, which sharpens rather than diminishes the contrast.

7.3 Open questions

(Q1) **The reopened target: a Lorentzian/Krein quantum-metric convergence theorem.** The degeneracy theorem collapses the former "weak-form vs strong-form" split into a single open problem, with a sharper specification supplied by the failure analysis.

Any viable construction must simultaneously: (i) use a temporal multiplier algebra the Dirac can see — Toeplitz compressions of multiplication operators e^{iqt} in the Connes–van Suijlekom S^1 pattern [6], not momentum-diagonal polynomials (else the temporal direction is metrically invisible, Lemma 4.4); (ii) not compress the Krein Dirac by P_+ — the compression annihilates the spatial part for every Krein-self-adjoint D_L of the product form (Theorem 5.2); candidate replacements include seminorms built from $J_L D_L$ or $|D_L|$ (Hilbert-space-self-adjoint surrogates), or a genuinely indefinite-inner-product Lipschitz analog; (iii) restore the kernel condition $L(a) = 0 \Leftrightarrow a \in \mathbb{C}1$ on whichever substrate is used — noting that the unrestricted natural substrate already fails it (Corollary 5.3), while chirality-flipping generators (Paper 46 Appendix B) are the natural starting point since their Dirac commutators are nonzero. Establishing the temporal Toeplitz commutator estimate of Lemma 4.10(d)'s caveat is the first concrete step. **Update (2026-06):** clauses (i) and (iii) are now closed in the action-seminorm framework, where the product-carrier convergence holds at additive rate (Proposition 5.9); clause (ii) is bypassed there, since the action seminorm never routes the metric through the Dirac. The remaining content of Q1 is therefore exactly the *signature*: a seminorm in which Lorentzian structure (not merely product topology) carries the metric data. The named candidate is a translation-type seminorm generated by the wedge boost / modular flow (Paper 42's four-witness identification, bit-exact at finite cutoff), so that causal structure enters through the generator. This remains a multi-month research target. A first finite-cutoff probe of the boost candidate (driver `debug/wh7_b3_boost_seminorm_probe.py`; frozen falsifier `tests/test_wh7_b3_boost.py`; Peter–Weyl window $j \leq 1$, full reachable multiplier system of rank 55, boost $K = 2J_z$ in Paper 42's doubled `two_m_j` convention) returns four exact structural facts: (a) $\sigma_{2\pi}(F) = F$ at 10^{-15} for every system element, and at *half* period the bit-exact band-parity grading $\sigma_\pi(F) = (-1)^{2b}F$ — a finite-cutoff spin–statistics shadow of the modular flow; (b) the boost-alone kernel is the *structured middle* between the two known failure modes: $\dim\ker(\mathrm{ad}_K) = 9 = \sum_{b\in\{0,1,2\}}(2b+1)$, exactly the boost-invariant multipliers — neither the total annihilation of Theorem 5.2 ($\dim 55$) nor the metric kernel condition ($\dim 1$); (c) the kernel condition is restored by the transversal frame ($\dim\ker = 1$ for the joint $\{J_x, J_y, J_z\}$ system); (d) the signed leg comparison $Q(F) = \|[J_z, F]\|^2 - \|[J_x, F]\|^2 - \|[J_y, F]\|^2$ induces a nontrivial causal classification of multiplier classes $(b, |m'|)$ whose signs agree with the symbol-level classifier $\mathrm{sign}(2m'^2 - b(b+1))$ in all nine classes: $m' = 0$ classes purely spacelike (Q-ratio $= -1$ exactly), top weights timelike from $b = 3/2$ upward, and the $b = 1$ top weight sitting *on the cone* ($q_{\min} = 0$ to 10^{-6}). Phase 2 (driver `debug/wh7_b3_phase2_cone_structure.py`; falsifier `tests/test_wh7_b3_phase2.py`) closes the classifier's closed form and the rate-level reverse-triangle question: (e) in Hilbert–Schmidt norm the causal ratio equals the symbol classifier *exactly*,

$$q_F\big(C^b_{m'm}\big) \;=\; \frac{2m'^2 - b(b+1)}{b(b+1)},$$

an exact rational for every system element (exact Clebsch–Gordan arithmetic; float cross-check 2×10^{-16}) — the window compression is *invisible* to the HS cone, and the null locus $2m'^2 = b(b+1)$ is exact; in operator norm the null ray is the single element $C^1_{\pm1,0}$ (residual 10^{-16}), with the remaining op-norm cone values simple rationals at float precision (3/11, 5/13, 3/5; closed-form proof open); (f) the causal form Q is class-diagonal with *definite* type per weight sector — inertia $(8, 0, 0)$ and $(10, 0, 0)$ on the timelike classes, identically zero on the six-dimensional $b = 1$ top class, negative-definite on the spacelike classes. In particular the timelike sector is positive-definite, so the rate functional $\tau = \sqrt{Q}$ is *sub*additive there and the operator-rate-level reverse triangle fails (by inertia, confirmed 120/120 in sampling): the cone is a **grading**, not a metric signature. The Lorentzian (reverse-triangle) content must therefore enter at the *state* level along modular orbits — thermal-time intervals with super-additivity from relative-entropy monotonicity, i.e. the Paper 49 cocycle machinery, which survives the descope — and that construction, together with any convergence statement, is the remaining (Phase 3) target. A first state-level panel (`debug/wh7_b3_phase3_state_intervals.py`; falsifier `tests/test_wh7_b3_phase3.py`) lays the foundation: wedge KMS state, flow-invariance, 2π orbit period and orbit injectivity all at 10^{-13} or better; the Datta max-divergence chain inequality (the universal "a detour never costs less" direction) verified 96/96 on kicked orbit triples; and a reproducible bimodal scaling of the kick-attributable cost with the kick generator's boost weight ($\sim \epsilon^2$ for $|m'| \leq 1/2$, $\sim \epsilon^{1.2}$ for $|m'| \geq 1$; not a reparametrization artifact — tangent projection shifts the exponents by < 0.1). Honest caveat: the *excess* over the on-orbit baseline is reference-state dependent (sign included); only the absolute chain inequality is universal. A second state-level panel (`debug/wh7_b3_phase3_sprint2.py`; falsifier `tests/test_wh7_b3_phase3_sprint2.py`) resolves both the scaling and the caveat by a single symmetry: when the kick generator commutes with the boost ($[G, K] = 0$, zero weight transfer — exactly the $m' = 0$ classes; the commutator norm equals the weight transfer $2|m'|$ identically), conjugation by $e^{i\epsilon G}$ commutes with the modular flow and unitary invariance of the cost gives the exact leg identity $c_{12}(\epsilon) = c_{23}(-\epsilon)$, so the kick excess is an *even* function of ϵ — quadratic leading order for *every* unitarily invariant cost functional (verified at 10^{-14}; the per-leg first-order terms do *not* vanish, they cancel between legs). For weight-transferring kicks the identity breaks at $O(\epsilon)$ and a signed linear term appears whose generic sign *is* the reference-state caveat; the apparent $\epsilon^{1.2}$ exponents were window fits of signed-linear-plus-quadratic mixtures. A five-functional comparison (max-divergence, Umegaki, Bures angle, trace distance, Jeffreys) over a 504-cell reference-state ensemble shows *no* cost functional attains state-independent positive excess (Bures closest, 0.93 overall and 1.00 on the three low-transfer classes on that panel — refuted as a general positivity by the third panel below), while the Umegaki and Jeffreys *chain* inequalities fail generically on modular-orbit triples (90/96 and 88/96 with negative baseline deficit; the Sprint-1 0/96 was reference-state luck, re-confirming the Datta max-divergence replacement of Paper 49). On the proper substrate — the hemispheric wedge with unfolded K_W (odd positive integer spectrum, genuine Boltzmann KMS state) — the anchors hold at 10^{-15}, the orbit period is π rather than 2π ($U_\pi = -\mathbb{1}$; attribution corrected by the third panel: this holds already on the *full* Dirac space at every cutoff, since the all-odd $2m_j$ spectrum makes weight differences even — the period-π is the spinor spin–statistics grading $\sigma_\pi(F) = (-1)^{2b}F$ restricted to a half-integer sector, *not* a folding effect), the commuting/non-commuting dichotomy transfers bit-level ($\Delta w = 0$ even at 10^{-13} with $p \approx 2$; $\Delta w \geq 2$ signed linear), and the interval functional ℓ = flow parameter is *operational*: recovered from state pairs alone by trace-distance matching, well-defined mod π and additive on ordered triples at 10^{-3} (grid-limited). The architecture this fixes: the interval is the flow parameter, costs enter as an even (quadratic) penalty layer for admissible (weight-conserving) deformations and a signed first-order layer otherwise. A third state-level panel (`debug/wh7_b3_phase3_sprint3.py`; falsifier `tests/test_wh7_b3_phase3_sprint3.py`) settles the admissibility and convergence questions at sprint grade. *Admissibility:* the wedge folding *reorganizes* the Phase-2 causal classes, and a fourth, exact-arithmetic panel (`debug/wh7_b3_phase3_sprint3b_fold_rule.py`; falsifier `tests/test_wh7_b3_fold_rule.py`) gives the closed form. Blockwise, plain-swap reflection conjugation acts as $[R\,C^b_{\mu'\mu}\,R]_{(j_1,j_2)} = (-1)^{b+j_2-j_1}\,[C^b_{-\mu',\mu}]_{(j_1,j_2)}$ (a Clebsch–Gordan symmetry, verified exactly on all class generators), so each (j_1, j_2) block of a $\mu = 0$-column Hermitized generator is an R-parity eigenblock with $\varepsilon = (-1)^{b+j_2-j_1+\mu'}$, and the folding annihilates a block iff $\varepsilon = -1$ (exact rational Frobenius^2 fold ratios: 6/19, 5/6, 0, 11/19, 1/2 on the $\mu = 0$ column; 3/8, 1/4 on the mixed classes). On the $j \leq 1$ window the $b = 2$ system has only the $(1, 1)$ block — which is *why* the $(2, 1)$ class is annihilated

($\varepsilon = -1$) and the folded $(2, 2)$ timelike class is purely mirror, hence flow-commuting (even penalty at 10^{-15}, median $|D^+|$ dropping from 0.26 upstairs to 3×10^{-4}). Both headline effects are *window-edge* facts: at $j_{\max} = 3/2$ the $(2, 1)$ class revives through the half-integer $\varepsilon = +1$ blocks (exact fold ratio 8/41, blocks $(1/2, 3/2)$ and $(3/2, 1/2)$) and the folded $(2, 2)$ class acquires non-mirror transitions and no longer commutes with K_W (mirror Frobenius fraction exactly 9/25). The invariant content: admissibility is the **flow-commutation grading**, not the causal grading; $\mu' = 0$ generators fold to weight-diagonal (admissible) operators at *every* tested window; and the mirror component (transitions $m'_1 = -m'_2$) is always admissible. Penalty magnitudes order by weight transfer (Spearman $r = 0.97$) at least as well as by the causal ratio ($r = 0.90$). *Convergence:* with band-limited reference perturbation and kicks ($n_{\text{fock}} \le 2$ labels, identical at every cutoff), every penalty and derivative datum is bit-exactly **cutoff-independent** across $n_{\max} = 2..5$ (deviations 10^{-15}–10^{-11}): the Gibbs normalization cancels in the $D_{\max}$ ratio operator and band-limited conjugations leave the bulk untouched. Meanwhile the naive trace-norm orbit scale decays exactly as $0.1805/Z$ — the finite-cutoff shadow of the type-III continuum wedge: only band-relative ($D_{\max}$-based) objects survive the limit, and the $D_{\max}$-matching interval recovery is cutoff-stable at 2×10^{-4}. The convergence statement for fixed band-limited data is therefore *exact*, not asymptotic; the genuinely open limit is band exhaustion (increasing bands). A fifth panel (multi-agent; shared substrate `debug/wh7_band_exhaustion_lib.py`, probes `debug/wh7_band_exh_*.py`; falsifier `tests/test_wh7_band_exhaustion.py`) closes that question at the numerical level on the folded Peter–Weyl wedge at $j_{\max} \in \{1, \ldots, 3\}$ (wedge dimensions 9–78) with fixed multiplier data. *The interval layer needs no limit:* operational recovery and additivity are bit-exact at every window ($\le 4\times10^{-16}$, no growth trend), and the flow-translation identity $c_{12} = c_{23}$ holds at 10^{-15} throughout. *The cost layer converges at its own rate:* generic-reference leg costs and deficits converge with strictly shrinking successive differences whose ratios (0.67–0.74, increasing) **exclude both** the metric-layer γ-rate (predicted ratios 0.93–0.95) **and** the thermal rate (predicted constant e^{-1}; independently excluded by the β-discriminator, which goes the wrong way by $\sim 10^3$, and by an adversarial $\beta = 0.5$ re-run that leaves the ratios essentially unchanged) — a power-law class with local exponent $p_{\text{eff}} \approx 1.5$, window-degenerate with rational $p \to 2$ forms (an internal counterexample with known $p = 2$ fits $p_{\text{eff}} = 1.48$ on this ladder), implying a state-level error rate $\sim (2j)^{-1/2}$ — *slower* than the metric layer: the two layers have genuinely different convergence modes. *The spin–statistics grading appears in the exhaustion dynamics:* integer-b data freezes bit-exactly across half-integer shell additions (the b-parity staircase, the $(-1)^{2b}$ grading of the boost-seminorm panel), and the null-class compression norm obeys the exact closed form $\|P_W M_{C^1} P_W\|_2 = \sqrt{6}\,(2j_{\max})/(2j_{\max} + 2)$ (deviation $\le 10^{-15}$ at every window; limit $\sqrt{6} = \sqrt{b(b+1)(2b+1)}$ at $b = 1$ — algebraic, π-free). Honest qualifications: non-commuting kick penalties do *not* stabilize monotonically across the ladder (and their monotonicity labels are reference-amplitude-dependent); the trace-scale$\times Z$ law of the fourth panel does NOT transport to this substrate; and the exponent identification ($p = 3/2$ vs rational $p \to 2$) is open pending more windows or the analytic form. A Mondino–Sämann-grade limit theorem remains future analytic work; these panels fix what it must prove. *Sign-definiteness:* the Bures even-sector positivity observed on the second panel is **refuted** by an adversarial panel of 2400 random flow-commuting kicks (574 negative cells, worst -2.3×10^{-2}; the second panel's own class generators go negative once the perturbation strength and flow time widen) — no tested functional has a sign-definite even-sector penalty; the chain inequality remains the only universal direction.

(Q2) **De-compactification limit $T \to \infty$.** The norm-resolvent (operator-level) arrow to $S^3 \times \mathbb{R}_t$ survives (Remark 5.8; Paper 47 [44] § 4). The metric-level statement on the non-compact carrier is open, downstream of Q1.

(Q3) **Cross-manifold extensions.** The cross-manifold extension $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ is blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry (Paper 24 [35] § V). The four-layer Coulomb/HO asymmetry there (spectrum-computing L_0, calibration π, Wilson gauge with natural matter, modular-Hamiltonian structure of the wedge KMS state) precludes the natural extension within the present framework.

(Q4) **Sharp constants, deferred behind Q1.** The rate expressions retain $C_3^{\mathrm{joint}} \leq 1$ with finite-cutoff form $\sup_{N \leq 2n_{\max}-1}(N-1)/\sqrt{N^2-1} = \sqrt{1-1/n_{\max}}$ (per Remark 4.5). On the momentum-diagonal substrate the temporal contribution to the Lipschitz seminorm is identically zero for *all* multipliers (not only pure tensors), so questions about joint sharpness are moot until Q1 supplies a construction in which the temporal direction carries metric weight.

(Q5) **Quantitative rate constants on the $U(1)$ factor.** The $U(1)$ leading constant in $\gamma^{U(1)} = O(T \log N_t/N_t)$ is standard Cesàro/Fejér; a Stein–Weiss-style derivation of a joint constant matching the SU(2) factor's $4/\pi$ becomes meaningful only within a Q1-repaired construction.

7.4 Structural-skeleton scope

The results here obey the structural-skeleton scope discipline of the GeoVac framework in its negative form: the annihilation theorem is a forced structural statement (Krein-self-adjointness plus compression leaves no freedom), and the spatial convergence statement determines rate expressions on the S^3 factor at finite cutoff. Neither determines calibration data (Yukawa selection, inner-factor parameters), which are external input. The selection of $T = 2\pi$ is structurally tied to the Bisognano–Wichmann modular period via Paper 43; the construction itself works for any $T > 0$.

Acknowledgments

The author thanks the developers of `cvxpy`, `numpy`, `scipy`, `sympy`, and `mpmath` for the computational infrastructure underlying the numerical verification. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude; implementation, internal sprint memos, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author's responsibility.

A Symbolic Plancherel calculation for the joint Fejér symbol

We record the kernel's two Plancherel-side objects. A caution worth stating explicitly: the identity $\int_{\mathrm{SU}(2)} K^{\mathrm{SU}(2)} \chi_{j'}\, dg = (2j'+1)/Z$ — which would make the character coefficients equal the mass distribution — is *false* (already at $n_{\max} = 2$, $j' = \frac{1}{2}$ the left side is $2\sqrt{2}/3 \neq 2/3$); the two objects are related but distinct, as follows.

Lemma A.1 (Mass distribution and multiplier symbol). *Let $K^{\mathrm{SU}(2)} = Z^{-1}|h|^2$ with $h = \sum_{j \leq j_{\max}} \sqrt{2j+1}\,\chi_j$ and $Z = Z^{\mathrm{SU}(2)}_{n_{\max}} = n_{\max}(n_{\max}+1)/2$, and let $K^{U(1)}$ be the Fejér kernel (11). Then, exactly over the relevant real quadratic extensions of $\mathbb{Q}$:*

(a) (Mass distributions.) *The squared Plancherel weights of the amplitudes give $m^{\mathrm{SU}(2)}(j) = (2j+1)/Z$ on $j \leq j_{\max}$ and $m^{U(1)}(k) = N_t^{-1}$ on $|k| \leq K_{\max}$; both are probability distributions, and the joint mass factorizes, $m^{K^{\mathrm{joint}}} = m^{\mathrm{SU}(2)} \cdot m^{U(1)}$.*

(b) (Multiplier symbols.) *Expanding* $K^{\mathrm{SU}(2)} = \sum_{j'}(c_{j'}/Z)\,\chi_{j'}$ *by the SU(2) Clebsch–Gordan rule* $\chi_{j_1}\chi_{j_2} = \sum_{J=|j_1-j_2|}^{j_1+j_2}\chi_J$*,*

$$c_{j'} \;=\; \sum_{\substack{j_1,j_2\le j_{\max}\\ |j_1-j_2|\le j'\le j_1+j_2\\ j_1+j_2+j'\in\mathbb{Z}}} \sqrt{(2j_1+1)(2j_2+1)},$$

and the convolution operator $f\mapsto K^{\mathrm{SU}(2)} * f$ *acts on the* j'*-isotypic block by the scalar*

$$\sigma^{\mathrm{SU}(2)}_{n_{\max}}(j') \;=\; \frac{c_{j'}}{(2j'+1)\,Z}, \qquad \sigma^{\mathrm{SU}(2)}(0)=1, \quad 0\le\sigma^{\mathrm{SU}(2)}\le 1, \quad \mathrm{supp}\subseteq\{j'\le 2j_{\max}\}.$$

At $n_{\max}=2$*:* $(c_0,c_{1/2},c_1) = (3,2\sqrt{2},2)$ *and* $\sigma^{\mathrm{SU}(2)} = (1,\sqrt{2}/3,2/9)$ *at* $j'=(0,\tfrac12,1)$*. For the* $U(1)$ *factor, direct computation gives* $\sigma^{U(1)}(k) = \max(0,(N_t-|k|)/N_t)$ *(Katznelson [15] Ch. I.2 at this normalization). The joint multiplier symbol factorizes,* $\sigma^{K^{\mathrm{joint}}} = \sigma^{\mathrm{SU}(2)}\cdot\sigma^{U(1)}$*, via Peter–Weyl on the direct product (characters multiply).*

Proof. (a) is immediate from the definition of the amplitudes (the weights $\sqrt{2j+1}$ and the flat Dirichlet window respectively), with normalization $\sum_j(2j+1)=Z$ and $(2K_{\max}+1)/N_t=1$. (b): $|h|^2 = \sum_{j_1,j_2}\sqrt{(2j_1+1)(2j_2+1)}\,\chi_{j_1}\chi_{j_2}$ (SU(2) characters are real), and the Clebsch–Gordan rule contributes each admissible j' once per pair — j' running in *unit* steps from $|j_1-j_2|$ to j_1+j_2, so only pairs with $j_1+j_2+j'\in\mathbb{Z}$ contribute; $\sigma(j') = c_{j'}/((2j'+1)Z)$ is the standard convolution eigenvalue on the j'-block for a central kernel with character coefficient $c_{j'}/Z$. Unitality at $j'=0$ is $c_0=\sum_j(2j+1)=Z$. Non-negativity and the upper bound follow since σ is the symbol of a positive unital map. The $n_{\max}=2$ values are direct, and are verified bit-exactly against independent Weyl-integration quadrature in the repository (`debug/p45_adversarial_L2_memo.md`). □

B Numerical verification panel data

The full numerical panel data tabulated in Table 1 is reproduced from the project repository (`debug/data/l3b_2_sub_sprint_D.json`); the corresponding computational driver is `debug/l3b_2_sub_sprint` and the production module is `geovac/lorentzian_propinquity_compact_temporal.py` (approximately 700 lines, with the `LorentzianTunnelingPair` dataclass, the `LorentzianPropinquityBound` dataclass, the `compute_lorentzian_propinquity_bound` function, and the `verify_riemannian_limit_at_N_t` bit-exact check).

The regression suite `tests/test_lorentzian_propinquity.py` (15+ test functions) verifies the formula-level content: the tunneling-pair construction, the assembly constituent inequalities of § 5.3 (classical-gradient ball), the rate-formula panel, the asymptotic ratio, the $\mathcal{K}^+$ compression algebra, the $N_t=1$ formula reduction, and the monotone decrease in $n_{\max}$. The operator-level degeneracy of § 5.2 is frozen as a separate falsifier in `tests/test_p45_kplus_degeneracy.py` (driver `debug/p45_kplus_seminorm_check.py`).

Convergence diagnostic. The ratio $\Lambda^L(4,7)/\Lambda^L(2,3) = 1.3223/2.0746 = 0.6374$ matches the single-factor Paper 38 / Paper 39 ratio bit-identically, confirming that the SU(2) factor dominates the joint rate at the tested cutoffs. At larger cutoffs the $U(1)$ factor would re-enter at order $O(T/N_t)$ if N_t does not grow as fast as $n_{\max}/\log n_{\max}$.

Memory note. The panel cell $(n_{\max},N_t)=(5,9)$ was originally planned but exceeds the operator-system vec-matrix memory budget at $\dim_K = 1260$ and $K = 2565$ multipliers (≈ 60 GiB `complex128`). The panel is restricted to $\{(2,3),(3,5),(4,7)\}$, which is sufficient for Riemannian-limit recovery, monotone-decrease, and asymptotic-rate consistency. Larger cells would require sparse linear-algebra refactoring of `geovac.operator_system_compact_temporal`, which is a downstream engineering item and not an analytical blocker.

References

[1] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan's Property (T)*, New Math. Monographs **11**, Cambridge University Press, 2008.

[2] J. J. Bisognano and E. H. Wichmann, *On the duality condition for quantum fields*, J. Math. Phys. **17** (1976), 303–321.

[3] N. Bizi, C. Brouder, and F. Besnard, *Space and time dimensions of algebras with application to Lorentzian noncommutative geometry and quantum electrodynamics*, J. Math. Phys. **59** (2018), 062303; arXiv:1611.07062.

[4] M. Bożejko and G. Fendler, *Herz–Schur multipliers and uniformly bounded representations of discrete groups*, Arch. Math. **57** (1991), 290–298.

[5] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Dirac operator on spheres and real hyperbolic spaces*, J. Geom. Phys. **20** (1996), 1–18; arXiv:gr-qc/9505009.

[6] A. Connes and W. D. van Suijlekom, *Spectral truncations in noncommutative geometry and operator systems*, Comm. Math. Phys. **383** (2021), 2021–2067; arXiv:2004.14115.

[7] A. Devastato, F. Lizzi, and P. Martinetti, *Time as an emergent property in spectral physics: gauge theories from spectral noncommutative geometry*, J. Math. Phys. **59** (2018), 112301.

[8] C. Farsi and F. Latrémolière, *Continuity for the spectral propinquity of the Dirac operators associated with an analytic path of Riemannian metrics*, arXiv:2504.11715, 2025.

[9] N. Franco and M. Eckstein, *An algebraic formulation of causality for noncommutative geometry*, Class. Quantum Grav. **30** (2013), 135007; arXiv:1212.5171.

[10] Y. Gaudillot-Estrada and W. D. van Suijlekom, *Convergence of spectral truncations for compact metric groups*, arXiv:2310.14733, 2023.

[11] R. Geroch, *Topology in general relativity*, J. Math. Phys. **8** (1967), 782–786.

[12] J. B. Hartle and S. W. Hawking, *Path-integral derivation of black-hole radiance*, Phys. Rev. D **13** (1976), 2188–2203.

[13] E. Hekkelman, *Truncated geometry on the circle*, arXiv:2111.13865, 2021.

[14] E. Hekkelman and E. McDonald, *A noncommutative integral on spectrally truncated spectral triples, and a link with quantum ergodicity*, J. Funct. Anal., to appear (2025); arXiv:2412.00628.

[15] Y. Katznelson, *An Introduction to Harmonic Analysis*, 3rd ed., Cambridge University Press, 2004.

[16] F. Latrémolière, *The Gromov–Hausdorff propinquity for metric spectral triples*, Adv. Math. **404** (2022), Paper No. 108393, 56pp; preprint arXiv:1811.10843, November 2018.

[17] F. Latrémolière, *The quantum Gromov–Hausdorff propinquity*, Trans. Amer. Math. Soc. **368** (2016), 365–411.

[18] M. Leimbach and W. D. van Suijlekom, *Gromov–Hausdorff convergence of spectral truncations of the torus*, Adv. Math. **439** (2024), Paper No. 109496.

[19] E. Minguzzi and S. Suhr, *Lorentzian metric spaces and their Gromov–Hausdorff convergence*, Lett. Math. Phys. **114** (2024), Paper No. 73; arXiv:2209.14384.

[20] A. Mondino and C. Sämann, *Lorentzian Gromov–Hausdorff convergence and pre-compactness*, arXiv:2504.10380, 2025.

[21] M. Che, R. Perales, and C. Sormani, *Gromov's compactness theorem for the intrinsic timed-Hausdorff distance*, arXiv:2510.13069, October 2025.

[22] G. Nieuviarts, *Emergence of Lorentz symmetry from an almost-commutative twisted spectral triple*, arXiv:2502.18105 (v3, May 2025).

[23] G. Nieuviarts, *Emergence of time from a twisted spectral triple in almost-commutative geometry*, arXiv:2512.15450, December 2025 (revised May 2026).

[24] V. I. Paulsen, *Completely Bounded Maps and Operator Algebras*, Cambridge Studies in Advanced Mathematics 78, Cambridge University Press, 2002.

[25] J.-P. Pier, *Amenable Locally Compact Groups*, Wiley, 1984.

[26] G. Pisier, *Similarity Problems and Completely Bounded Maps*, Lecture Notes in Math. 1618, Springer, 2nd ed., 2001.

[27] G. L. Sewell, *Quantum fields on manifolds: PCT and gravitationally induced thermal states*, Ann. Phys. (NY) **141** (1982), 201–224.

[28] A. Strohmaier, *On noncommutative and pseudo-Riemannian geometry*, J. Geom. Phys. **56** (2006), 175–195.

[29] M. Takesaki, *Theory of Operator Algebras I*, Springer, 1979.

[30] R. Toyota, *Quantum Gromov–Hausdorff convergence of spectral truncations for groups with polynomial growth*, arXiv:2309.13469, 2023.

[31] W. G. Unruh, *Notes on black-hole evaporation*, Phys. Rev. D **14** (1976), 870–892.

[32] K. van den Dungen, *Krein spectral triples and the fermionic action*, Math. Phys. Anal. Geom. **19** (2016), Art. 4, 22pp; arXiv:1505.01939.

[33] W. D. van Suijlekom, *Gromov–Hausdorff convergence of state spaces for spectral truncations*, J. Geom. Phys. **162** (2021), 104075; arXiv:2005.08544.

[34] J. de Groot, *Pseudo-Riemannian Spectral Triples for* $\mathrm{SU}(1,1)$, arXiv:2601.22171, January 2026.

[35] J. Loutey, *Bargmann–Segal lattice for the 3D harmonic oscillator and the Coulomb/HO structural asymmetry*, GeoVac internal preprint (2026), DOI via Zenodo.

[36] J. Loutey, *The GeoVac spectral triple, the master Mellin engine case-exhaustion theorem, and the GH-convergence theorem on* S^3, GeoVac internal preprint, papers/synthesis (2026), DOI via Zenodo.

[37] J. Loutey, *Latrémolière propinquity convergence of truncated Camporesi–Higuchi spectral triples on* S^3, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.

[38] J. Loutey, *Tensor-product propinquity convergence for distinct-focal-length truncated Camporesi–Higuchi spectral triples on* S^3, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.

[39] J. Loutey, *Latrémolière propinquity convergence of spectral truncations of compact Lie groups with bi-invariant metric*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.

[40] J. Loutey, *Tomita–Takesaki modular structure on truncated* SU(2) *spectral triples; four-witness Wick-rotation literal identification at finite cutoff*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.

[41] J. Loutey, *Lorentzian extension of the four-witness Wick-rotation theorem at finite cutoff*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.

[42] J. Loutey, *Operator-system extension of the BBB Krein spectral triple at finite cutoff: propagation number and envelope dependence on the Camporesi–Higuchi spinor bundle*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo.

[43] J. Loutey, *Strong-form Lorentzian propinquity convergence on truncated SU(2)* $\times$ *U(1)*$_T$ *Krein spectral triples*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo. (Companion paper closing the strong-form gap identified as Q1/G1 of the present paper; identified the envelope refinement to (21) in Remark 4.5.)

[44] J. Loutey, *Norm-resolvent convergence to the non-compact Lorentzian Krein spectral triple on* $S^3 \times \mathbb{R}_t$ *and the two-rate hybrid limit*, GeoVac internal preprint, papers/standalone (2026), DOI via Zenodo. (Closes the de-compactification limit G2 of the present paper at the norm-resolvent / spectral level via a two-rate hybrid composite; propinquity / metric level on a non-compact carrier remains open as named G2-metric / G1-genuine.)